General Mathematics

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Learning Strategies

Mathematics is often the most challenging subject for students. Much of the trouble comes from the fact that mathematics is about logical thinking, not memorizing rules or remembering formulas. It requires a different style of thinking than other subjects. The students who seem to be "naturally" good at math just happen to adopt the correct strategies of thinking that math requires – often they don't even realise it. We have isolated several key learning strategies used by successful maths students and have made icons to represent them. These icons are distributed throughout the book in order to remind students to adopt these necessary learning strategies:



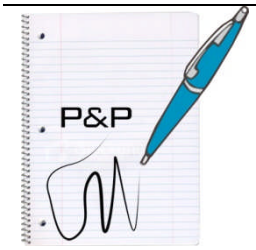
Talk Aloud Many students sit and try to do a problem in complete silence inside their heads. They think that solutions just pop into the heads of 'smart' people. You absolutely must learn to talk aloud and listen to yourself, literally to talk yourself through a problem. Successful students do this without realising. It helps to structure your thoughts while helping your tutor understand the way you think.



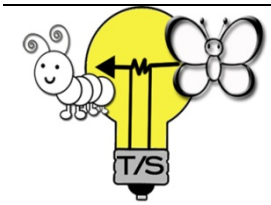
BackChecking This means that you will be doing every step of the question twice, as you work your way through the question to ensure no silly mistakes. For example with this question: $3 \times 2 - 5 \times 7$ you would do "3 times 2 is 5 ... let me check – no 3×2 is 6 ... minus 5 times 7 is minus 35 ... let me check ... minus 5×7 is minus 35. Initially, this may seem time-consuming, but once it is automatic, a great deal of time and marks will be saved.



Avoid Cosmetic Surgery Do not write over old answers since this often results in repeated mistakes or actually erasing the correct answer. When you make mistakes just put one line through the mistake rather than scribbling it out. This helps reduce silly mistakes and makes your work look cleaner and easier to backcheck.



Pen to Paper It is always wise to write things down as you work your way through a problem, in order to keep track of good ideas and to see concepts on paper instead of in your head. This makes it easier to work out the next step in the problem. Harder maths problems cannot be solved in your head alone – put your ideas on paper as soon as you have them – always!



Transfer Skills This strategy is more advanced. It is the skill of making up a simpler question and then transferring those ideas to a more complex question with which you are having difficulty.

For example if you can't remember how to do long addition because you can't recall exactly how to carry the one: $\begin{array}{r} 5889 \\ +\,4587 \\ \hline \end{array}$ then you may want to try adding numbers which you do know how to calculate that also involve carrying the one: $\begin{array}{r} 5 \\ +\,9 \\ \hline \end{array}$

This skill is particularly useful when you can't remember a basic arithmetic or algebraic rule, most of the time you should be able to work it out by creating a simpler version of the question.

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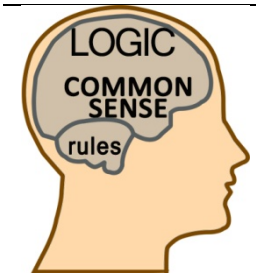
Format Skills These are the skills that keep a question together as an organized whole in terms of your working out on paper. An example of this is using the "=" sign correctly to keep a question lined up properly. In numerical calculations format skills help you to align the numbers correctly.

This skill is important because the correct working out will help you avoid careless mistakes. When your work is jumbled up all over the page it is hard for you to make sense of what belongs with what. Your "silly" mistakes would increase. Format skills also make it a lot easier for you to check over your work and to notice/correct any mistakes.

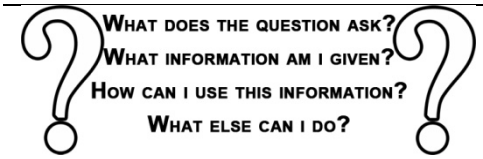
Every topic in math has a way of being written with correct formatting. You will be surprised how much smoother mathematics will be once you learn this skill. Whenever you are unsure you should always ask your tutor or teacher.



Its Ok To Be Wrong Mathematics is in many ways more of a skill than just knowledge. The main skill is problem solving and the only way this can be learned is by thinking hard and making mistakes on the way. As you gain confidence you will naturally worry less about making the mistakes and more about learning from them. Risk trying to solve problems that you are unsure of, this will improve your skill more than anything else. It's ok to be wrong – it is NOT ok to not try.



Avoid Rule Dependency Rules are secondary tools; common sense and logic are primary tools for problem solving and mathematics in general. Ultimately you must understand Why rules work the way they do. Without this you are likely to struggle with tricky problem solving and worded questions. Always rely on your logic and common sense first and on rules second, always ask Why?



Self Questioning This is what strong problem solvers do naturally when they get stuck on a problem or don't know what to do. Ask yourself these questions. They will help to jolt your thinking process; consider just one question at a time and Talk Aloud while putting Pen To Paper.

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Table of Contents

CHAPTER 1: Financial Mathematics **5**

Exercise 1: Earning Money 6

Exercise 2: Taxation 10

Exercise 3: Credit & Borrowing 15

Exercise 4: Annuities & Loan Repayments 19

Exercise 5: Depreciation 22

CHAPTER 2: Data Analysis **25**

Exercise 1: Data Collection & Sampling 26

Exercise 2: Mean, Median & Spread of Data 30

Exercise 3: Representing Data (I) 34

Exercise 4: Representing Data (II) 39

Exercise 5: Normal Distribution 45

Exercise 6: Correlation 48

CHAPTER 3: Measurement **51**

Exercise 1: Units of Measurement 52

Exercise 2: Applications of Area & Volume 57

Exercise 3: Similarity 64

Exercise 4: Right Angled Triangles 69

Exercise 5: Further Applications of Trigonometry 78

Exercise 6: Spherical Geometry 86

CHAPTER 4: Probability **89**

Exercise 1: Simple Probability 90

Exercise 2: Multi-stage Events 96

Exercise 3: Applications of Probability 99

CHAPTER 5: Algebraic Modeling **103**

Exercise 1: Algebraic Skills & Techniques 104

Exercise 2: Modelling Linear Relationships 109

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General Mathematics

Financial Mathematics

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Exercise 1

Earning Money

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1) Mark earns a gross salary of $78000 per annum. To the nearest cent how much does Mark earn:

a) Per month

b) Per fortnight

c) Per week

d) Per day

e) Per hours

Assume Mark works a 40 hour week, does not work weekends, and ignore public holidays

2) Tom earns a gross salary of $900 per 37 hour week. Matt earns $22 per hour, but is required to work for 42 hours per week.

a) Who earns more per hour?

b) Who earns more per week?

c) What is the difference in their annual earnings? (Assume they each work for all 52 weeks of the year, not on weekends, and ignore public holidays)

3) Peter's pay rates for a week's work are as follows

- $25 per hour for the first 35 hours
- $40 per hour for each hour worked over 35 hours
- An extra $2.50 per hour for each hour worked over 40 hours

Calculate Peter's earnings before tax for the following scenarios

a) Worked 32 hours

b) Worked 35 hours

c) Worked 43 hours

d) Worked 60 hours

4) When James takes holidays he is allowed a 7.5% extra on top of his holiday pay. James' salary is currently $82500. If he takes two weeks holiday, how much will he be paid for this period?

5) Ronald works as a car salesman. He gets paid a base wage of $900 per week. He also gets paid commission for every car he sells, according to the sale price. If the car is valued below $20000 he gets 1% of the sale price. For cars sold in the $20000 to $39999 price range, he receives 1.5% commission. If the value of the car sold is $40000 or more he receives 2%.

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What does Ronald earn per week under the following scenarios?

a) He sells no cars

b) He sells one car valued at $32000

c) He sells a car for $35000 and one for $41950

d) He sells 4 cars all for $37500

If Ronald wanted to earn $2000 for a week's work, what must he sell a luxury car (valued at over $40000) for?

6) Petra dyes flowers and gets paid 1.5 cents for every stem she dyes.

a) If she dyes 3000 stems how much does she earn?

b) If she dyes 15000 stems, how much does she earn?

c) How many stems must she dye in order to earn $750?

7) New start allowance is paid to unemployed job seekers. A single person receives $492.60 per fortnight, whilst a couple receives $444.70 each per fortnight. A job seeker with a dependent child receives $533 per fortnight.

A carers pension is paid to anybody caring for a disabled child and pays $115.40 per fortnight

The aged pension is $712 per fortnight for a single pensioner and $536.70 each per fortnight for a married couple

Calculate how much each household brings in under the following conditions

a) Bill and Doris are both old aged pensioners, and their son Malcolm is currently seeking work

b) Jill is seeking work and also cares for her 10 year old son who is not disabled

c) Bob is a single pensioner who shares a house with his grandson John who is seeking work and also cares for his own son who has a disability

8) Bernard worked 37 hours last week. His hourly rate is $31.50, and he pays tax at a flat rate of 15% of his earnings. In addition he pays 1.5% of his gross pay toward the Medicare levy, and he also has to pay 4.5% of his gross pay in HECS repayments. Union fees of $8 and social club fees of $2.50 per week are also deducted.

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Bernard makes voluntary superannuation contributions of 3% of his gross pay.

How much money did Bernard actually take home last week?

9) Max works a 37 hour week and is paid for all public holidays also. He has the following weekly financial commitments

- Rent $350
- Electricity $35
- Petrol $50
- Gas $25
- Entertainment $75
- Food etc. $125
- Credit card $18
- Car costs $30

Max also wishes to put money away for such things as clothing, furniture, household items etc. so that he can pay cash for them when he needs them. He estimates he will need $1500 for the year.

Max also wishes to save $40 per week.

What must Max's hourly pay rate be to be able to meet his commitments and savings needs? (Assume Max does not pay taxation nor has any other deductions from his wages)

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Exercise 2

Taxation

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1) Martin works for a salary of $52000 per annum before tax. The weekly tax on this income is $162.44. How much does Martin take home per fortnight?

2) Income between $18201 and $37000 per annum is currently taxed at the rate of 19 cents per dollar for amounts over $18200. How much tax is payable for the following incomes?

a) $19200

b) $26000

c) $36999

d) $50000

e) $15000

3) People earning over $180000 per annum pay tax according to the following formula. $54547 plus 45 cents per dollar for each dollar over $180000. How much tax is payable for the following incomes?

a) $190000

b) $225000

c) $500000

d) $100000

The rates mentioned in questions 2 and 3 are taken from the following table which shows the formula to calculate tax payable on all incomes. Use the table to answer the following questions

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Taxable income	Tax on this income
0 - $18,200	Nil
$18,201 - $37,000	19c for each $1 over $18,200
$37,001 - $80,000	$3,572 plus 32.5c for each $1 over $37,000
$80,001 - $180,000	$17,547 plus 37c for each $1 over $80,000
$180,001 and over	$54,547 plus 45c for each $1 over $180,000

4) What is the annual tax payable for the following incomes?

a) $39125

b) $125432

c) $12000

d) $37000

e) $180002

f) $1,000,000

5) Jim earns $42 per hour for a 38 hour week. How much tax should be deducted from his wages each week to meet his taxation commitment?

6) Graph tax payable per annum versus taxable income for incomes from $0 to $200000

7) The Medicare levy is payable by all taxpayers who earn more than $20542 per annum, and is charged at the rate of 1.5% of taxable income. How much Medicare levy is payable for the following incomes?

a) $42222

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b) $17000

c) $82000

d) $53149

8) If an unmarried taxpayer is not covered by private health cover and they earn more than $84000 per annum, they are liable for the Medicare levy surcharge, which is a further 1% of taxable income

What is the total levy (including surcharge if applicable) payable for the following incomes?

a) $2000

b) $73250

c) $83999

d) $92000

e) $113000

9) Alan is single, and earned $93450 in the past financial year. His employer deducted $500 per week to cover his tax and Medicare commitments. At the end of the financial year is Alan due a refund from the government, or is he liable for additional tax?

10) GST is a tax placed on many items by the government; it is added to the base price of the item and is included in the total cost of the item. The current rate of GST is 10%. What is the total cost of the following items with base prices of:

a) $1.50

b) $12.50

c) $105.00

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d) $32000

e) $12243.56

11) Use guess check and improve, or develop a method to calculate the base price of the following items that have a total cost of:

a) $11

b) $44

c) $36.19

d) $111.32

e) $8938.05

Develop a formula that enables you to calculate the base price of an item given its total cost

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Exercise 3

Credit & Borrowing

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1) Calculate the total simple interest paid under the following conditions

a) Principal of $10,000 at a rate of 10% p.a. for 10 years

b) Principal of $2000 at a rate of 5% p.a. for 5 years

c) Principal of $4000 at a rate of 7.5% p.a. for 2 years

d) Principal of $25,000 at a rate of 12.5% p.a. for 3 years

e) An interest rate of 8% p.a. for 5 years on a principal of $6,000

2) Calculate the amount of time it would take to repay a loan under the following conditions (assume simple interest)

a) Principal of $5,000 at 10% p.a. interest with total interest payable of $2000

b) Principal of $12,000 at 12% p.a. interest with total interest payable of $6000

c) Principal of $2,000 at 20% p.a. interest with total interest payable of $6400

d) Principal of $800 at 11% p.a. interest with total interest payable of $440

3) A man borrows $11500 to buy a car. He agrees to a simple interest rate of 6% per annum and agrees to pay the loan off in 5 years. How much will he repay in total?

4) Kerry borrows $4000 and is required to repay the loan with equal monthly instalments. If the simple interest rate is 9% p.a. how much will she have to repay each month to finalise the loan in 3 years?

5) A man takes out a loan of $10000 at 6.5% p.a. simple interest rate for 4 years. After 2 years the interest rate was increased to 8%. How much did his repayments have to increase by to still have the loan repaid in the same time?

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6) Complete the following table

Home loan table				
	Amount = $100,0000	Assume the same number of days per month		
	Interest Rate = 15% p.a.			
	Monthly repayment = $3000			
N	**Principal**	**Interest**	**P+I**	**P+I-R**
1	100000	1250	101250	98250
2	98250			
3				
4				
5				

7) From the table above, what would the amount owing be after 5 months if the monthly repayment was doubled? Why is this amount not equal to half the amount owing after 5 months in question 6?

8) Tom buys a new lounge suite for $2400 using the store's credit facility. The store offers a two year non-interest period. After that time the interest charged on the outstanding balance is 18% p.a. simple interest payable monthly.

a) If Tom wishes to avoid any interest charges, what is the minimum amount per month he should pay?

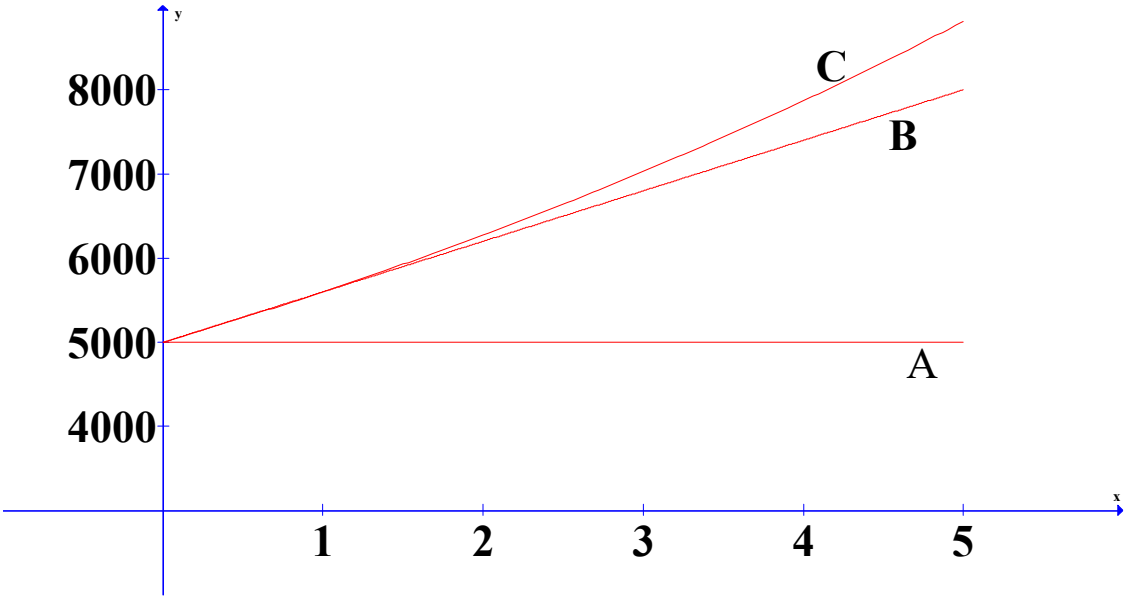
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b) If Tom repays the loan after 3 years with equal instalments, how much did he repay each month?

c) The store has a policy that if no repayments have been made in the first 30 months, the debt is referred to a collection agency. How much gets referred to the agency?

9) Which of the following curves represents

- The amount paid on a $5000 loan that is repaid with a simple interest rate
- The amount paid on a $5000 loan with a compound interest rate
- The amount paid on a $5000 loan repaid with no interest rate



10) Calculate the effective interest rate on a loan of $8000 at 15% p.a. interest paid monthly for 3 years

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Exercise 4

Annuities & Loan Repayments

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1) What is the future value of an annuity with a contribution of $100 per year for 15 years, if the interest rate is 10% p.a.?

2) What is the future of an annuity with a contribution of $2000 per 6 months for 20 years if the interest rate is 8% p.a.?

3) The future value of an annuity after 15 years is $80,000. If the interest rate was 20% p.a. what were the yearly contributions?

4) The future value of an annuity after 30 years is $250,000. If the interest rate was 9% p.a. and the contributions were made monthly, how much were these contributions?

5) Which has a greater future value; an annuity of $100 per month at 6% p.a. interest, or an annuity of $300 per quarter at the same interest rate? Assume the period of investment is 20 years, and explain why the two are not equal even though $100 per month is equal to $300 per quarter

6) Colin is saving for a place in a retirement village. If he needs $200,000 by the time he retires in 10 years, how much should he pay into an account each year if the rate of interest paid is 8% per annum?

7) John is planning to take the trip of a lifetime in ten years' time and estimates that the amount of money he will need at that time is $50 000. He is advised to contribute $4000 each year into an account that pays 5% pa, compounded annually. Will John have enough money in ten years time to make his dream come true? By how much will he fall short of or overshoot his goal?

8) What is the present value of an annuity of $150 per month @ 18% p.a. compounded monthly?

9) Peter has two options when saving for his retirement. Either invest $50000 today at 7% p.a. interest compounded annually for 10 years or pay $400 per month commencing immediately at 9% p.a. interest compounded monthly. Which option gives Peter more money to retire with?

10) In 8 years time a business plans to replace its fitting and fixtures. It is estimated that the replacement will cost $15000. How much does the business need to save per year if it receives 6% p.a. compounded annually on their savings?

11) Arnold deposits $200 per month into his account. How much does he have in his account at the end of 5 years if the bank pays 8% p.a.

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interest compounded every 2 months?

12) A couple take a home loan of $250000 over 30 years at 12% p.a. compounded monthly. What are the monthly repayments, total amount paid, and total interest paid over the course of the loan?

13) Use the table below to calculate the value of an ordinary annuity of $200 per month which is invested at 4% per month for 4 months

Future values of $1

	Interest rate				
Period	**1%**	**2%**	**3%**	**4%**	**5%**
1	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500
3	3.0301	3.0604	3.0909	3.1216	3.1525
4	4.0604	4.1216	4.1836	4.2465	4.3101
5	5.1010	5.2040	5.3091	5.4163	5.5256
6	6.1520	6.3081	6.4684	6.6330	6.8019
7	7.2135	7.4343	7.6625	7.8983	8.1420
8	8.2857	8.5830	8.8923	9.2142	9.5491

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Exercise 5

Depreciation

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1) Assuming straight line depreciation, what is the financial life of the assets having a depreciation rate of?

a) 10%

b) 8.5%

c) 20%

d) 12.5%

e) 5%

2) What is the depreciation rate of an asset that has the following financial life? (Assume straight line depreciation)

a) 5 years

b) 20 years

c) 12 years

d) 25 years

e) 10 years

3) A car with a book value of $50,000 is bought by a business in July 2006. If its value is depreciated by 20% using the straight line method, what is its book value in July 2010?

4) In July 2003 a computer system was valued at $8000. In July 2006 its value was $5000. Assuming straight line depreciation what was the depreciation rate?

5) A car originally bought for $40,000 was depreciated using the reducing balance method at a rate of 12%. What was its value after 1, 2 and 3 years?

6) In July 2006 office furniture was bought for $18000. It was depreciated using the reducing balance method, and in July 2009 its value was $13122. What rate of depreciation was used?

7) In July 2001 a car having a value of $35000 was purchased. It was depreciated at a rate of 10% using the straight line method. When did the value of the car equal zero?

8) In July 2001 a car having a value of $35000 was purchased. It was depreciated at a rate of 10% using the reducing balance method. When did the value of the car equal zero?

9) A boat having a value of $75000 was purchased and it was depreciated at a rate of 15% using the reducing balance method

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a) Write a formula that calculates the value of the boat after one year

b) Write a formula that calculates the value of the boat after 2 years

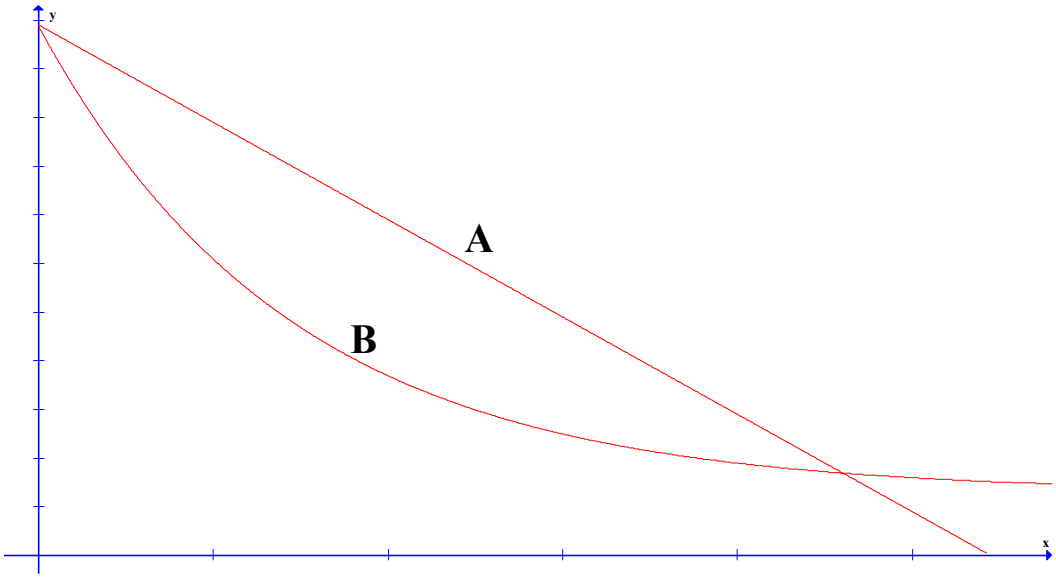
c) Write a formula that calculates the value of the boat after 5 years

d) Write a formula that calculates the value of the boat after n years

10) A car having a value of V dollars was purchased and then depreciated at a rate of 10% using the reducing balance method. Write a formula that could be used to calculate the value of the car after n years

11) A car having a value of V dollars was purchased and hen depreciated at a rate of r%. Write a formula that could be used to calculate the value of the car after n years

12) Which of the graphs below represents the value of an asset depreciated using the reducing balance method of depreciation? Explain your answer



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General Mathematics

Data Analysis

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Exercise 1

Data Collection & Sampling

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1) For which of the following would all data be available for analysis, and which would require a sample to be taken?

a) Score distribution in a basketball competition

b) Voting intentions of the Australian people

c) Favourite colour of your class

d) Favourite car of the people of Sydney

e) Types of dogs owned by the people of Victoria

2) Classify the following data as either quantitative or categorical. If the data is quantitative, indicate if it is discrete or continuous

a) Heights of your class members

b) Attendance at football games

c) Car colours

d) Dog breeds

e) Courses offered at a university

f) Number of people enrolled in each course at a university

3) Describe the differences and similarities between the random, stratified and systematic methods of sampling

4) A company employs workers under various conditions

- 50 workers are males who work full time
- 25 are males who work part time
- 75 are females who work full time
- 100 are females who work part time

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If stratified sampling is to be used, how many of each group should be sampled under the following conditions?

a) 50 people are to be surveyed in total

b) 25 females are to be surveyed

c) 75 part time workers are to be surveyed

d) 10 male part time workers are to be surveyed

5) The population of Australia is approximately 23 million. Of that number approximately 1,955,000 are over 65 years old. To gain an accurate representation of a sample set of 5000, how many of them should be over 65 years old?

6) A sample of 5000 people included 100 in the age range 20 to 40. Comment on the appropriateness of the sample distribution, given that the survey conducted related to services for parents of school aged children.

7) Tom made a table of the numbers of boys and girls in each year group in his school

YEAR	BOYS	GIRLS
1	12	15
2	9	14
3	13	12
4	9	10
5	16	15
6	11	14
7	12	17
8	14	17
9	13	15
10	9	11
11	8	10
12	6	8

Based on his data, approximately how many of the students in Tom's state are female? (The total number of students in Tom's state is 1,120,000)

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8) Peter also made a table of the number of boys and girls in each year group in his school

YEAR	BOYS	GIRLS
1	15	0
2	19	0
3	23	0
4	29	0
5	26	0
6	31	0
7	22	0
8	14	0
9	13	0
10	9	0
11	8	0
12	6	0

Comment on the suitability of using Peter's data for the same purpose as Tom's, the probable reason for its unsuitability, and what the data could possibly be used to estimate

9) 100 animals are caught, tagged and released. Later 250 animals are caught, of which 50 have tags. Based on this data what is the approximate population of these animals?

10) Based on tagging data, the population of fish in a lake is estimated to be 1000. Of the sample of 300 taken, 45 had tags already placed by a previous catch and release. How many fish were originally tagged and released?

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Exercise 2

Mean, Median & Spread of Data

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1) Calculate the mean of the following data sets

a) 2, 4, 6, 8, 10

b) 0, 2, 4, 6, 8

c) 1, 3, 5, 7, 9

d) 2, 2, 2, 2, 2

e) 10, 30, 40, 50

f) 7, 11, 15, 17, 25, 52, 55

2) Calculate the mean of the following data sets

a) 2, 4, 5, 7, 8

b) 2, 4, 5, 7, 8, 500

c) 950, 970, 990, 1000, 1100

d) 2, 950, 970, 990, 1000, 1100

3) From your answers to question 2, what effect does an outlier have on the mean of a set of data?

4) The mean of a set of data is 15. The scores in the data set are

18, 3, 15, x, 30, 12, and 20

What is the value of x?

5) Fifteen students sat a maths test and their mean mark was 60%. Alan was sick for the test and sat it later. When his score was added to the data set, the mean mark had increased to 62%. What score did Alan get on the test?

6) There are 15 girls and 15 boys in a class. On a test the girls mean mark was 80%, while the mean mark of the boys was 70%. What was the mean mark for the class?

7) There are 20 girls and 10 boys in a class. On a test the girls mean mark was 80% while the mean mark of the boys was 70%. What was the mean mark for the class?

8) Why are the answers to questions 6 and 7 different, given that the mean marks of the boys and girls in both classes were the same?

9) What is the median of the following data sets?

a) 1, 2, 3, 4, 5

b) 2, 4, 6, 8, 10

c) 9, 12, 15, 22, 30, 40, 60

d) 2, 4, 6, 12, 14, 21, 22, 22

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10) What is the median of the following data sets?

a) 2, 4, 5, 7, 10

b) 2, 4, 5, 7, 10, 1000

c) 1000, 982, 979, 977, 960

d) 1000, 982, 979, 977, 960, 2

11) From your answers to questions 10 and 11, what effect does an outlier have on the median of a set of data?

12) The following set of data is in order. Its mean is 30 and its median is 14. What are the values of x and y?

5, 8, x, 12, y, 40, 50, 100

13) Find the range of the following sets of data

a) 1, 2, 5, 7, 10

b) 3, 6, 18, 19, 100

c) 1, 1, 1, 1, 1

d) 17, 3, 18, 22, 30, 4, 10

e) 40, 30, 20, 10, 0

f) -5, 7, 15, 22, 40, 51

14) Find the inter-quartile range of the following data sets

a) 7, 15, 20, 22, 25, 32, 40

b) 1, 5, 6, 12, 20, 30, 50

c) 2, 10, 18, 24, 32, 80, 82, 90

d) 23, 25, 4, 12, 21, 50, 32, 43, 5, 60, 45

15) Can the inter-quartile range be less than the range for a set of data? Explain

16) Can the inter-quartile range be equal to the range for a set of data? Explain

17) What is the standard deviation of the following sets of data?

a) 2, 2, 2, 2, 2, 2

b) 1, 2, 3, 4, 5

c) 3, 6, 9, 12, 15

d) 4, 20, 40, 60, 100

18) Calculate the mean and standard deviation of the following

a) 2, 4, 6, 8, 10

b) 4, 6, 8, 10, 12

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c) What effect does adding two to every score have on the mean and standard deviation of a set of data?

19) Calculate the mean and standard deviation of the data set

4, 8, 12, 16, 20

What effect does doubling every score have on the mean and standard deviation of a set of data?

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Exercise 3

Representing Data (I)

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1) Create a tally chart and frequency table to represent the following data set more effectively

5, 7, 10, 16, 20, 6, 17, 9, 14, 4, 11, 12, 1, 2, 19, 14, 19, 10, 2, 15, 12, 17, 5, 1, 11, 13, 9, 7, 4, 8, 7, 3, 6, 16, 4, 1, 8, 5, 18, 13, 19, 9, 2, 11, 17, 17, 14, 10, 16, 4, 13, 1, 11, 15, 6, 3, 2, 7, 20, 8, 15, 6, 8, 5, 3, 11, 4, 10, 9, 13, 12, 18, 2, 17, 1

2) Construct a frequency histogram for the following grouped frequency table

Height of trees (metres)	**Frequency**
1 – 1.25	25
1.25 - 1.5	30
1.5 – 1.75	20
1.75 – 2	40
2 – 2.25	15
2.25 – 2.5	10
2.5 – 2.75	5

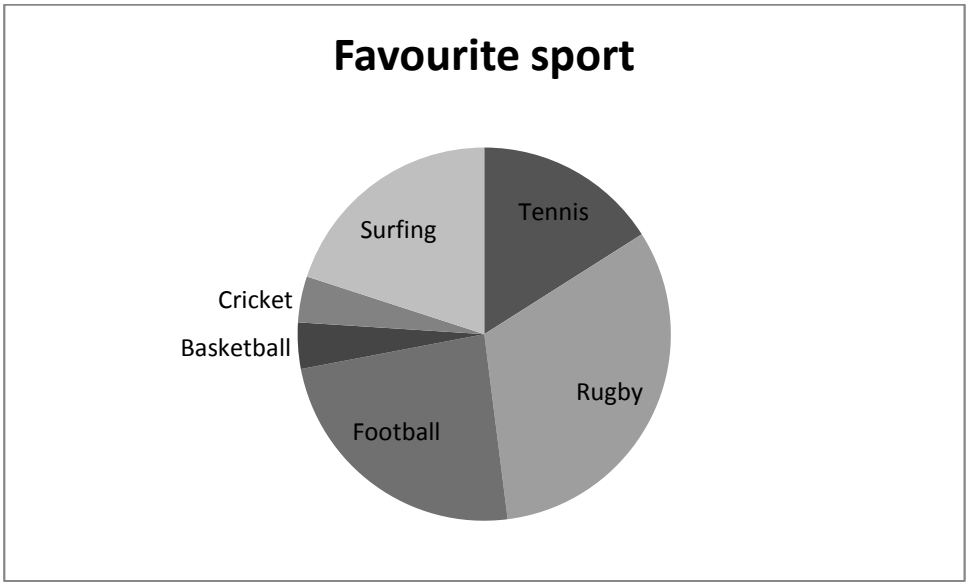
3) Construct a cumulative frequency table and graph for the data from question 2

4) Construct a pie graph to represent the following data

Hours of TV watched per week	**Number of people**
0-10	14
10-30	32
30-50	39
50-75	9
75+	6

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5) Using the following pie graph



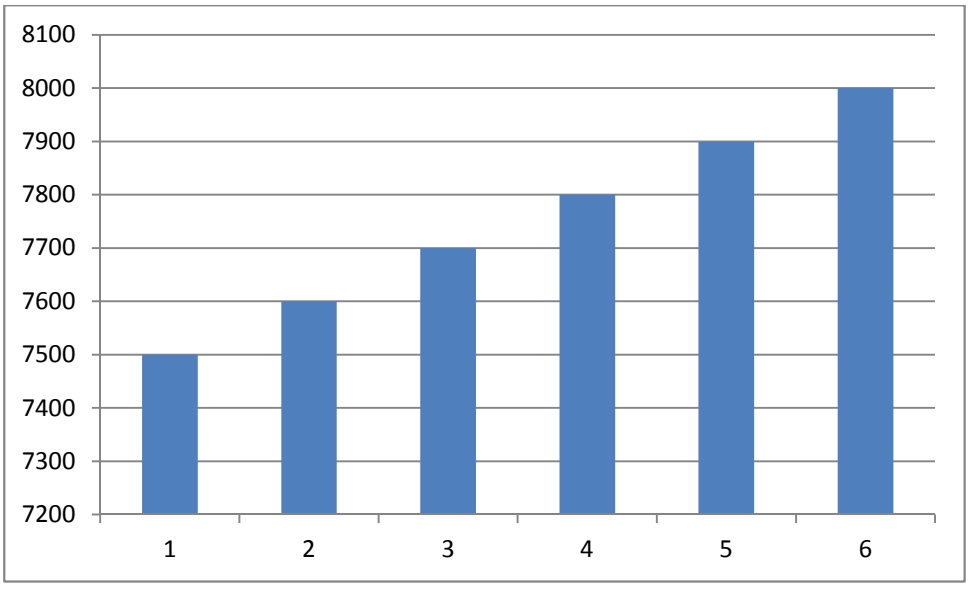
a) Which sport was most popular of those surveyed?

b) Which two sports were equally popular?

c) Which sport was the favourite of half the number of people who voted for rugby?

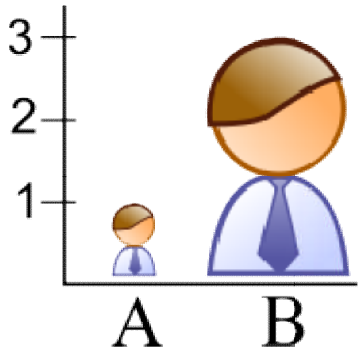
d) If 50 people chose surfing, approximately how many people were surveyed?

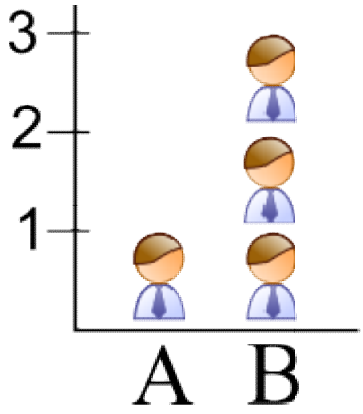
6) Explain why the following graph is misleading, and redraw it so as to make it realistic



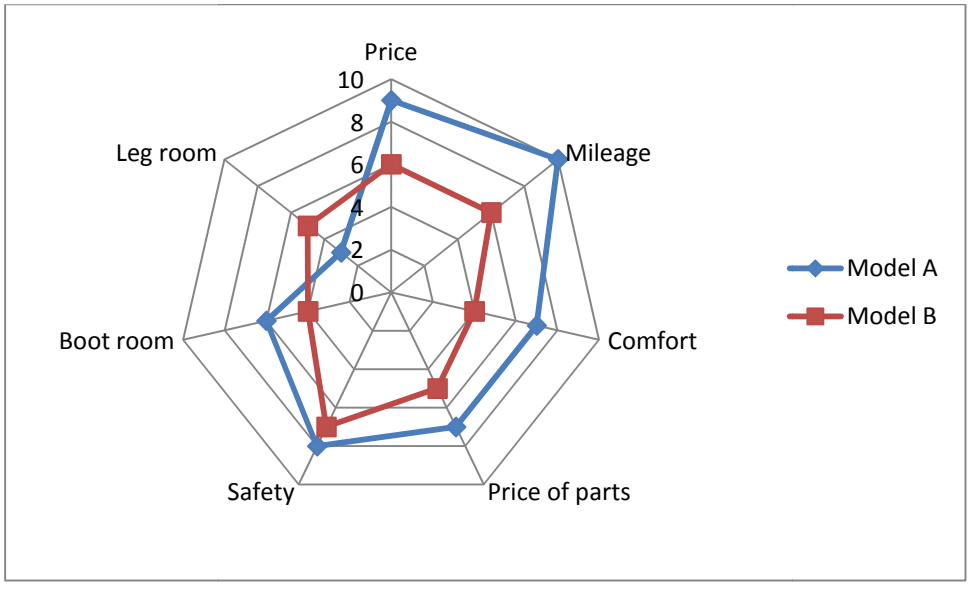
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7) Which of the picture graphs shown below is less misleading and why?





8) A magazine compared two cars named A and B in 7 criteria. The higher the score, the better the value. For example a high price score indicates that a car is cheaper, whilst a high safety score indicates that a car is safer



a) Which car is cheaper and by what fraction?

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b) Which car has more leg room?

c) Which feature scored almost the same for both cars?

d) What was the only category in which car B performed better than car A?

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Exercise 4

Representing Data (II)

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1) Represent the following data set in a stem and leaf plot and determine the median score using the plot

14, 15, 16, 16, 22, 23, 23, 23, 23, 24, 26, 31, 32, 38, 39, 44, 44, 45, 46, 47, 47, 47, 48

2) The daily maxima for Perth during the month of June 2012 were

19, 20, 22, 24, 23, 23, 17, 20, 21, 21, 19, 21, 20, 17, 18, 19, 18, 21, 24, 21, 16, 16, 17, 18, 19, 15, 21, 20, 19, 17,

Represent this data in a stem and leaf plot.

What was the median maximum temperature in Perth for June?

3) The following data set is the set of scores of football team A during its season

34, 38, 42, 43, 45, 48, 49, 51, 53, 57, 58, 60, 61, 63, 67, 71, 74, 77, 79, 85

The following data set is the set of scores of football team B during its season

23, 29, 35, 39, 46, 47, 49, 52, 53, 53, 59, 67, 73, 79, 86, 91, 97, 101, 117, 126

Display the data in a back to back stem and leaf plot

What were the respective median scores, and which team was more consistent during the season

4) Represent the following data set in a box and whisker plot

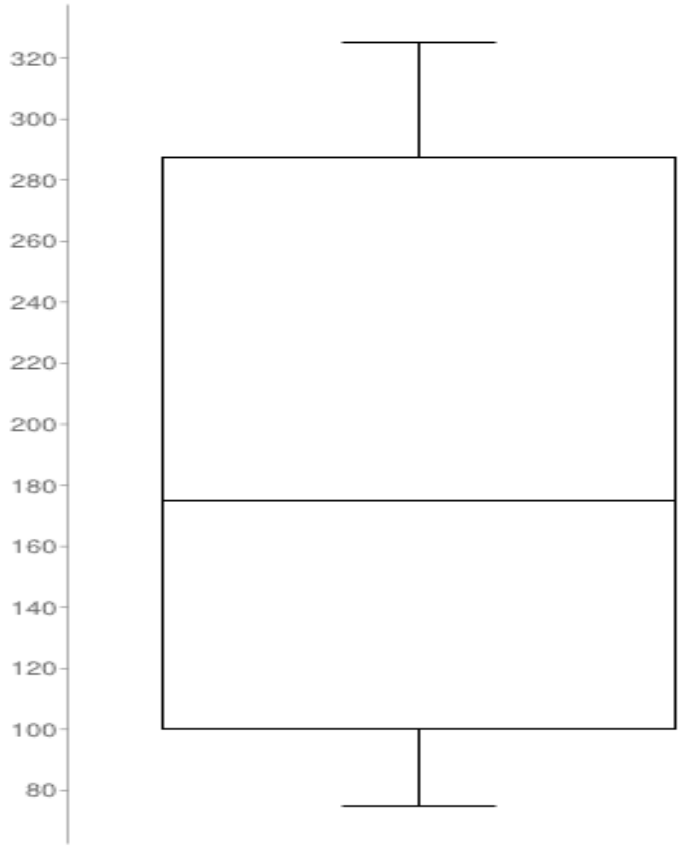
12, 16, 20, 24, 25, 30, 40, 42, 100

Show and evaluate the range and the inter-quartile range

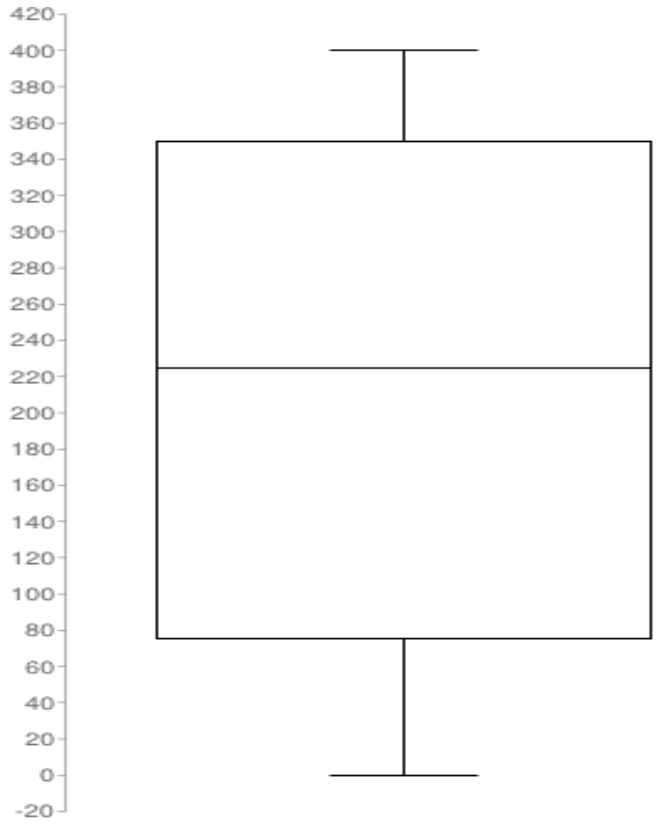
5) A set of data has a minimum of 4, an inter-quartile range of 15; range of 26 and a third quartile of 25. Draw a possible box and whisker plot for this data

6) The following box plot shows the distribution of the average rainfall for Great Lake for the past 40 years

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The following box plot shows the same data set for Water World



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a) Which site has the greater median average rainfall?

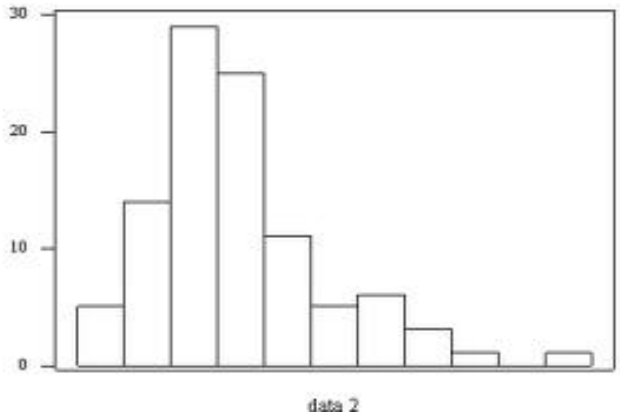
b) Which site has the record lowest annual rainfall and record highest annual rainfall?

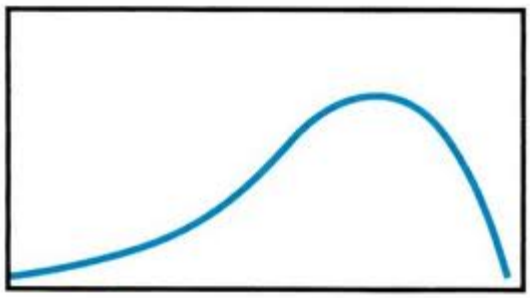
c) Which site has the greater variation in average rainfall?

d) Which site has a greater chance of receiving 300 inches or more of rain?

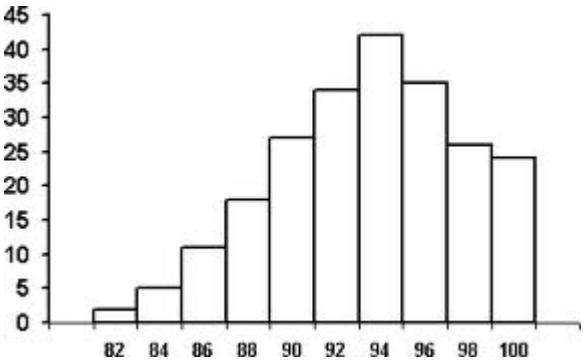
e) Too much or too little rain affects the water levels in the dam to the point where water skiing is too dangerous. Which site would give a person a better chance of being able to water ski?

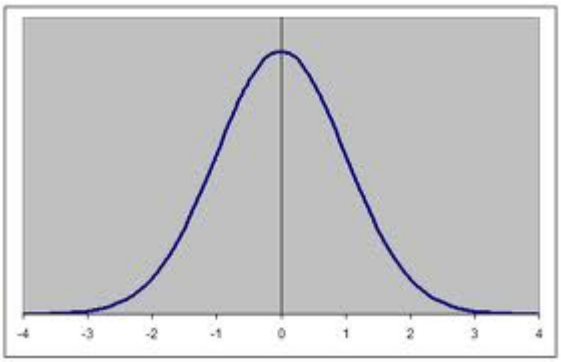
7) Describe the following graphs in terms of skewness



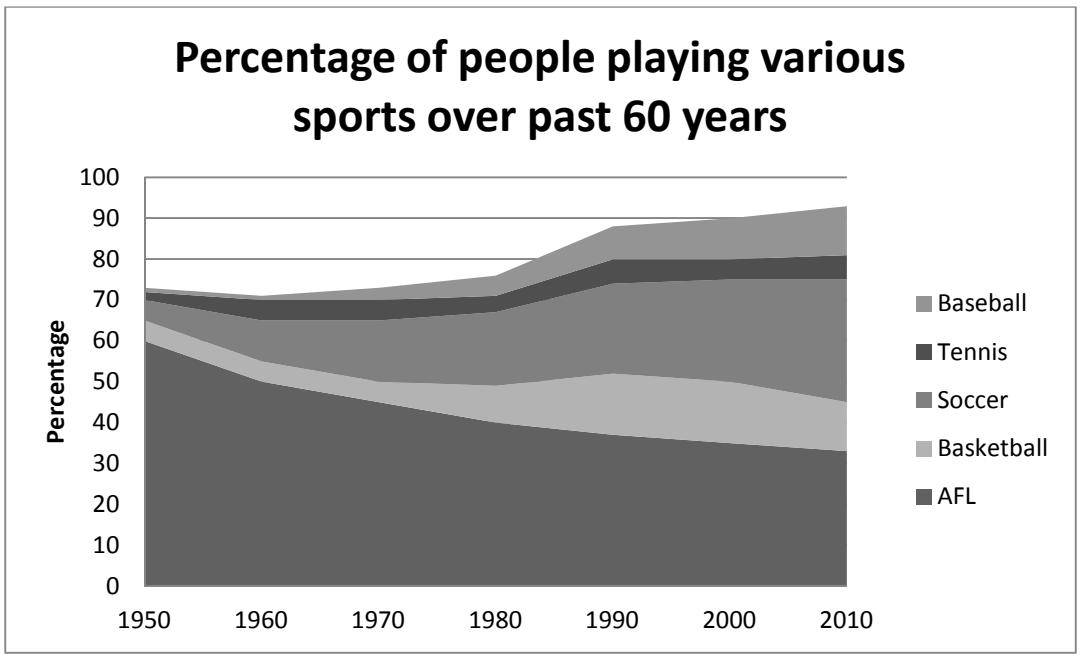


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8) Answer the questions below by using the following area graph



a) Which sport has had a steady decline in percentage participation rates?

b) To which sport has most of this percentage gone to?

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c) Which sport had the most rapid increase in participation percentage in the 1980s?

d) During which year was the total participation in these sports combined the highest?

e) Has the number of people playing AFL fallen over the past 60 years? Explain your answer.

f) The participation rate for which sport has remained relatively constant?

9) Answer the questions based on the following table

	Studied for test	**Did not study for test**	
Passed test	80	20	100
Failed test	10	90	100
	90	110	

a) What percentage of students passed the test?

b) What percentage of students who studied for the test passed it?

c) What percentage of students who did not study for the test failed?

d) If you failed the test what is the chance that you did not study?

10) 500 people were asked their preferred colour from red and blue. There were 150 women, 100 of whom liked blue. 200 men preferred red. What percentage of men preferred blue?

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Exercise 5

Normal Distribution

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1) Describe what the following z values tell us about the data point in relation to the mean

a) $z = 0$

b) $z = 1$

c) $z = -2$

d) $z > 2$

2) Calculate the z score of a score of 8 in a data set that has a mean of 6 and a standard deviation of 2. Describe the position of the data point in relation to the mean

3) A data point has a z score of 1.5. The data set has a mean of 5 and a standard deviation of 3. What is the data point?

4) A data set has a mean of 17.5. The data point 33.5 is 1.6 standard deviations from the mean. What is the value of the standard deviation?

5) The data point 41 lies within a set of data having a standard deviation of 6. If the data point is 4 standard deviations from the mean, what is the value of the mean?

6) If a set of data is normally distributed what percentage of the scores are within 1 standard deviation from the mean?

7) 95% of people in a group are between 77kg and 103 kg. What is the mean and standard deviation if we assume the data is normally distributed?

8) A teacher gives a maths test with the pass mark being 25 out of 50. The class scores the following marks:

12, 14, 10, 22, 35, 38, 13, 22, 40, 11, 22, 24, 25, 30, 5, and 18

The teacher sees that the majority of the class will fail the test, and he decides to standardise the marks. He will only fail a student that is more than one standard deviation below the mean

How many students now pass the test?

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9) Another teacher is determining the term marks for his class and wants to grade according to the following formula

Standard Deviations from mean	Grade
Score ≥2 s.d.	A
1 s.d. ≤ score < 2 s.d.	B
0 s.d. ≤ score < 1 s.d.	C
-1 s.d. ≤ score < 0 s.d.	D
Score< -1 s.d	E

Grade the following students

NAME	SCORE
James	62
Mark	38
Karen	84
Janine	70
Carol	65
June	68
Peter	44
Kevin	48
Brian	56
Alan	66
Bree	53

10) Deliveries of sand made by a nursery are advertised as 100 kg. The mean of the deliveries is 100 kg with a standard deviation of 1.2 kg

a) Within what weight range will 95% of the deliveries be?

b) What percentage of deliveries will be between 100 kg and 101.2 kg?

c) The company offers money back if any of the deliveries are 3 or more standard deviations below the mean. If they made 5000 deliveries in one month, how many of these will have to be refunded?

(Assume the data is normally distributed)

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Exercise 6

Correlation

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1) Plot the following sets of ordered pairs on their own scatter plot

a) $(1,2)\ (2,5)\ (3,7)\ (4,8)\ (5,12)\ (6,9)$

b) $(3,4)\ (6,11)\ (7,7)\ (9,30)\ (11,22)\ (12,35)$

c) $(10,12)\ (9,9)\ (8,4)\ (7,8)\ (6,10)\ (5,1)$

d) $(20,8)\ (14,12)\ (10,7)\ (7,10)\ (3,1)\ (2,5)$

e) $(20,2)\ (10,15)\ (3,7)\ (8,4)\ (5,2)\ (6,17)$

f) $(4,12)\ (2,6)\ (3,9)\ (1,3)\ (5,15)\ (6,18)$

2) For each set of data points in question 1, describe the relationship between the points as strong/medium/weak and positive/negative. Also indicate if any relationship is perfect or there is no relationship at all.

3) For any set of data from question 1 for which there is a relationship, draw the line of best fit through the data, and determine the gradient and vertical intercept. Hence determine the equation of the line of best fit

4) For each of the equations derived in question 3, predict the y value obtained when substituting the point $(3, y)$ into the equation

5) Explain why you could not predict the y value of the point $(40, y)$ in any of the equations above

6) Describe the relation between the two variables of a scatter plot that have the following correlation coefficients

a) $r = 1$

b) $r = 0.8$

c) $r = -0.1$

d) $r = 0.6$

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e) $r = -0.85$

f) $r = 0.09$

7) When the relationship between the sale of blankets in Canada and the sale of air conditioners in Australia at different times of a year is graphed in a scatter plot, the correlation coefficient for the line of best fit is 0.8. Does this mean that the number of air conditioners bought in Australia affects the number of blankets bought in Canada? Explain your answer

8) A scatter plot was produced that showed the relationship between the average life expectancy and the number of television sets per person for a number of countries. The correlation coefficient was very high $(r = 0.92)$. Does this mean that in order to increase life expectancy in third world countries, simply introduce more television sets? Explain your answer

9) Describe the likely scatter plot between the ages and heights of a randomly selected group of 5000 people. What do you think the value of the correlation coefficient may be, and are there any restrictions on the validity of the correlation coefficient? Explain your answer

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General Mathematics

Measurement

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Exercise 1

Units of Measurement

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1) Convert the following to cm

a) 8 mm

b) 1.5 m

c) 0.3 km

d) 412 mm

e) 22.65 m

f) 0.025 km

2) Convert the following to m^2

a) 4900 cm^2

b) 0.04 km^2

c) 320000 mm^2

d) 0.005 km^2

e) 22250 cm^2

3) Brian uses a ruler marked in centimetres to measure the lengths of various lines. What is the percentage error for each of the following measurements?

a) 400 cm

b) 12 cm

c) 2 m

d) 1200 mm

e) 0.3 km

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f) 3000 cm

4) Convert the following to metres per minute

a) 3 km per second

b) 10000 mm per hour

c) 1500 m per day

d) 20 km per hour

e) 525.6 km per year

5) The concentration of an additive in a solution is 1:500000. How much additive is present in the following amounts of solution?

a) 1 kg

b) 800 g

c) 10 kg

d) 0.6 kg

e) 10000 g

f) 300 kg

6) The concentration of an additive in a solution is 1 mg per 750 ml. How much additive is there in the following volumes?

a) 2 litres

b) 500 ml

c) 3 litres

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d) 20 litres

e) How much solution is there if it contains 12 g of additive?

7) What percentage of the original quantity remains after the following additions and reductions occur?

a) There is an increase of 10% then a decrease of 10%

b) There is a decrease of 10% followed by an increase of 10%

c) There is an increase of 50% followed by a decrease of 50%

d) There is an increase of 100% followed by a decrease of 100%

e) Does the answer change if the decrease occurs before the increase?

f) Develop a formula to calculate the above changes in one step, and validate it by checking it against the answer for a 20% decrease followed by a 20% increase.

8) The recommended dosage of a medicine is 5 ml plus an extra 1.5 ml per kg of weight of the patient over 50kg. What dosage should be given to patients with the following weights?

a) 41 kg

b) 103 kg

c) 75 kg

d) 30 kg

e) If a patient was given 20 ml of the medicine, what was their weight?

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9) Two powders (A and B) are to be mixed in the ratio 3:5. How much of powder A must be added to the following quantities of powder B?

a) 1.5 kg

b) 600 g

c) 10 kg

d) 200 mg

e) 1.4 g

f) 1000 kg

10) Solve the following

a) A mixture to make 12 cakes needs 300g of sugar, how much sugar is needed to make 16 cakes?

b) A car requires 65 litres of fuel to travel 800 km, how much fuel does it need to travel 900 km?

c) A plate of radius 10 cm holds 30 biscuits laid flat. What is the radius of a plate that holds 8 biscuits?

d) 15 cats require a total of 2.25 kg of food per day. How much food is needed for 35 cats in 2 days?

e) In 6 minutes a train travels 25 km. If its speed is constant, how far will it travel in 11 minutes?

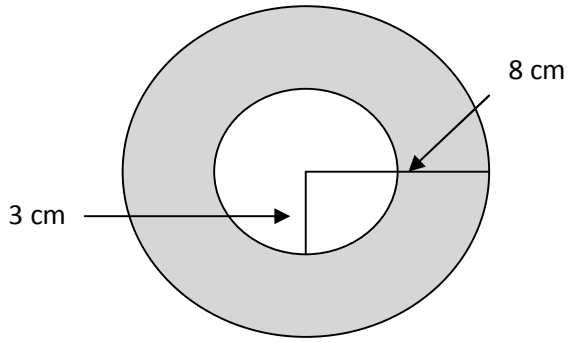
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Exercise 2

Applications of Area & Volume

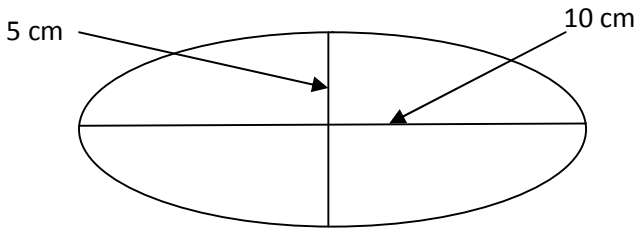
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1) Calculate the area of the annulus

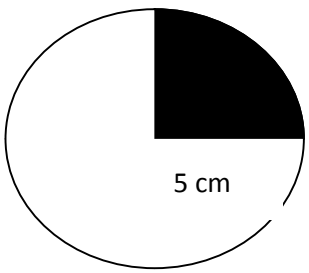


2) If the radius of the larger circle from question 1 is halved, and the radius of the smaller circle is doubled, what is the change in the area of the new annulus formed?

3) Calculate the area of the following figure

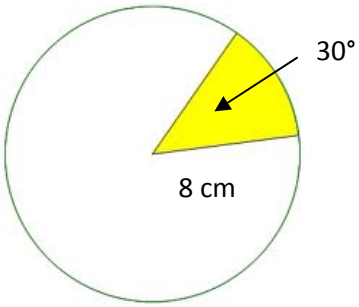


4) Calculate the shaded area



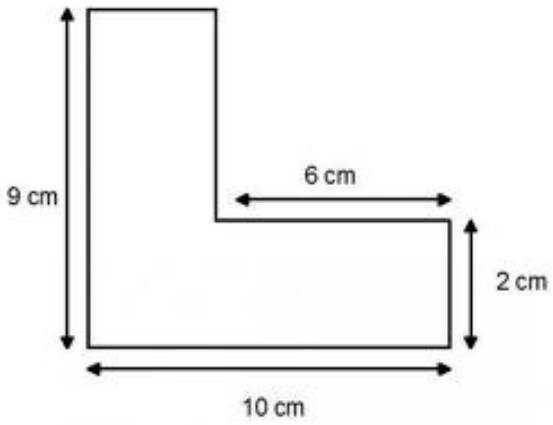
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5) Calculate the shaded area

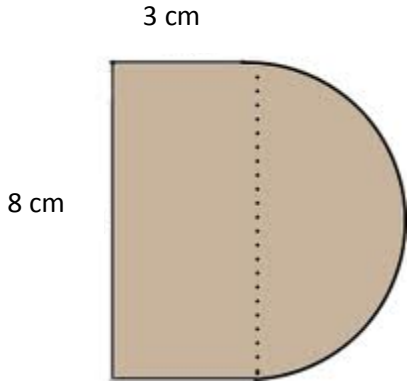


For questions 6 – 9, calculate the total area of each composite shape

6)

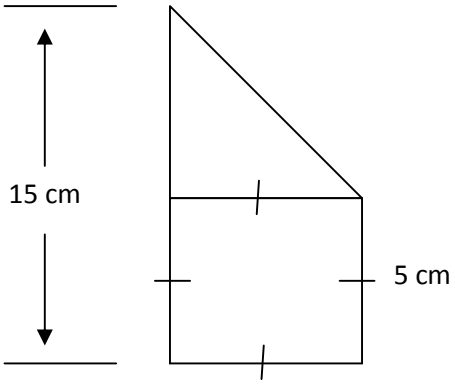


7)

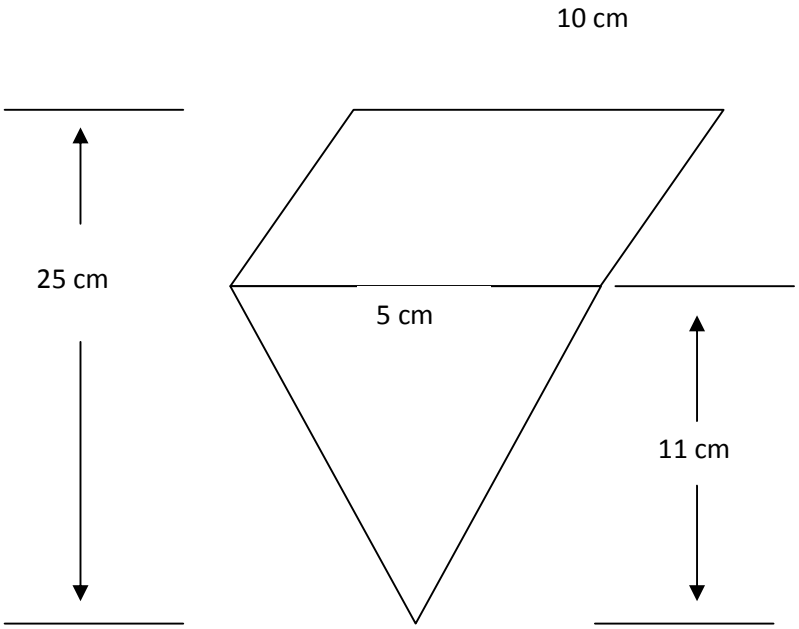


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8)

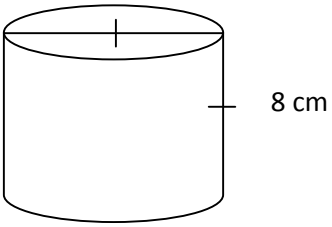


9)



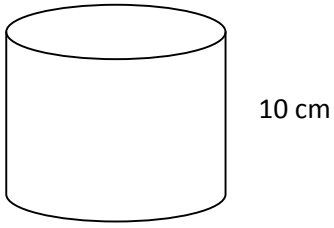
10) Calculate the surface area of the following cylinders (parts c and d are open cylinders; they have no top or bottom)

a)

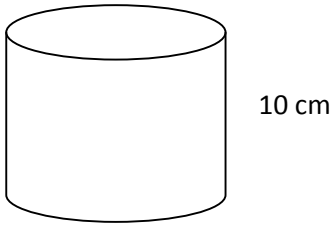


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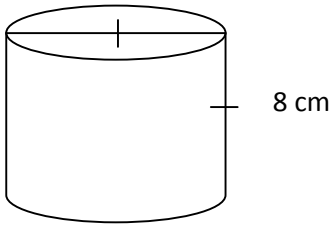
b) $r = 5\ cm$



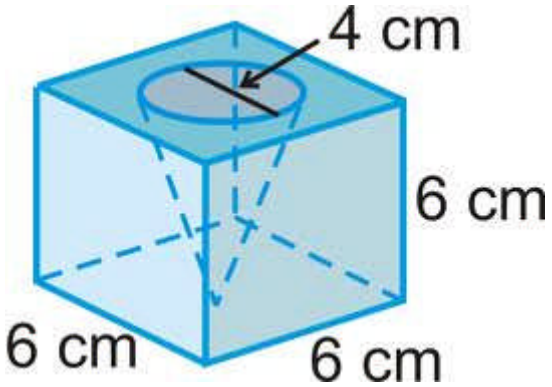
c) $r = 5\ cm$



d)



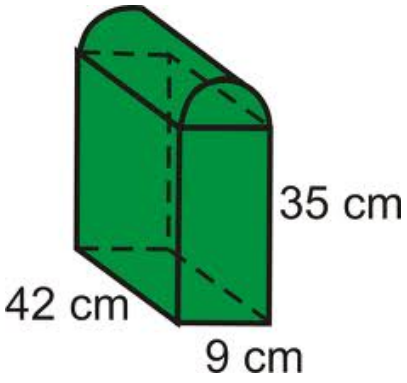
11) What is the total surface area of the following solid, which is a cube with a conic section cut out?



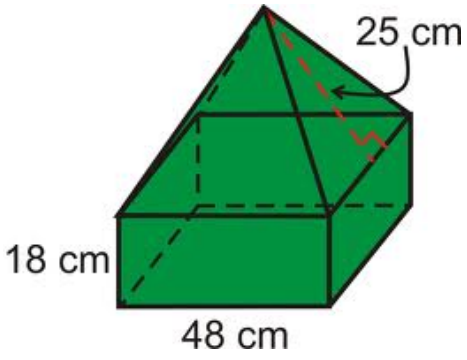
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12) Calculate the volume of the following solids

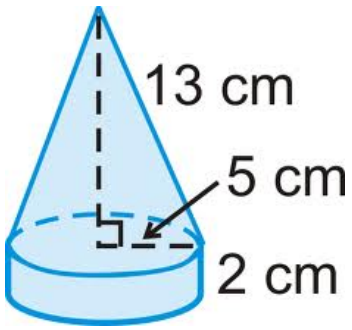
a)



b)

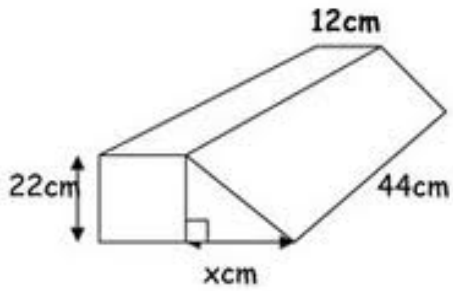


c)



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13) The volume of the solid below is 16456 cm^3. What is the value of x?



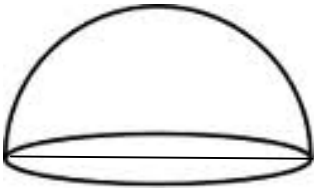
14) Calculate the surface area of a sphere with the following radii

a) 4 cm

b) 6 cm

c) 10 cm

15) Calculate the total surface area of the shape below



12 cm

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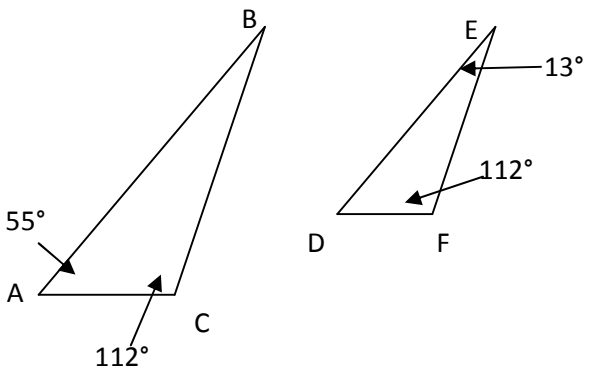
Exercise 3

Similarity

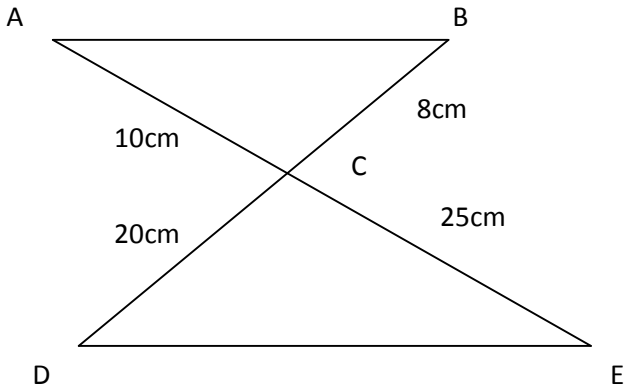
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1) Determine if each pair of triangles is similar. If so, state the similarity conditions met

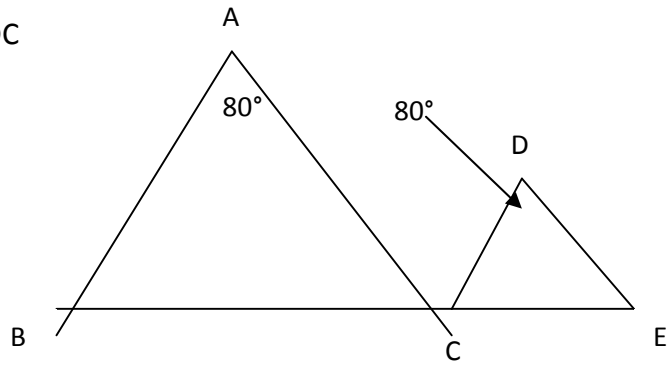
a)



b)

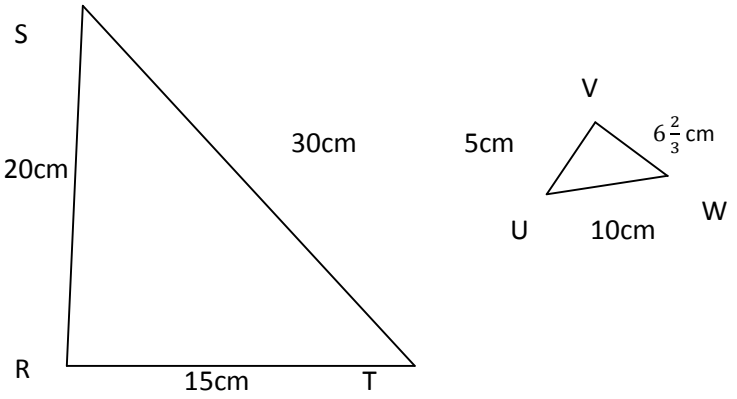


c) AB || DC

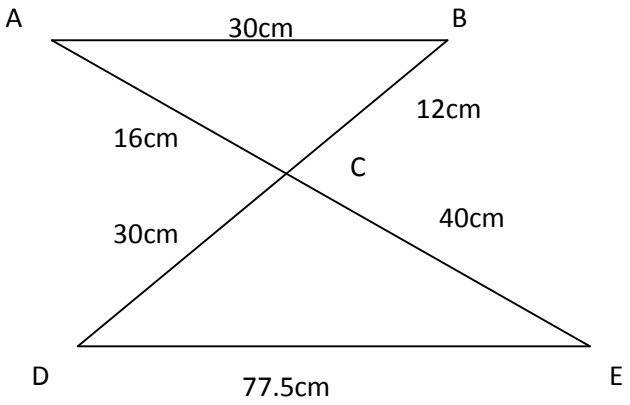


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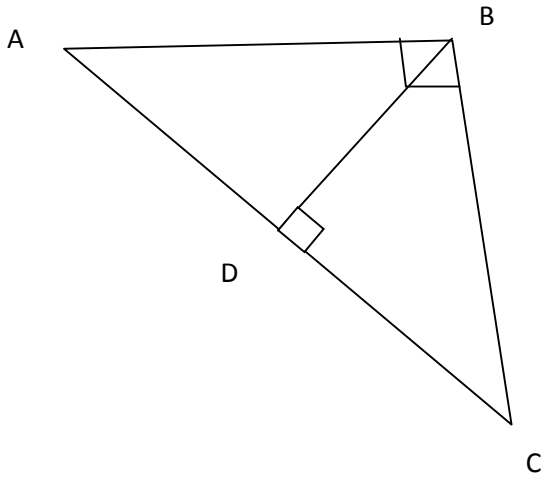
d)



e)

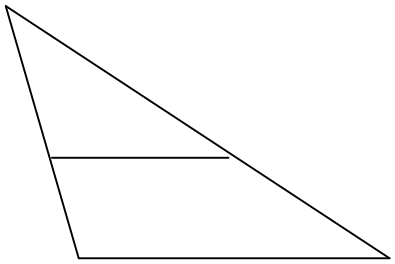


f)

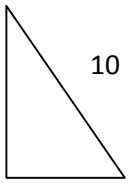


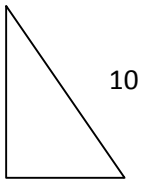
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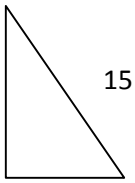
2) What additional information is needed to show that the two triangles are similar by AAA?



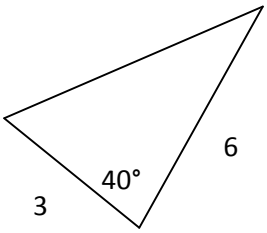
3) Of the following three right-angled triangles, which two are similar and why?

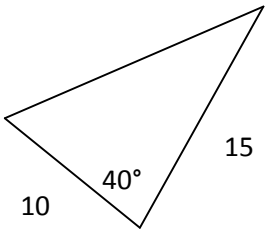


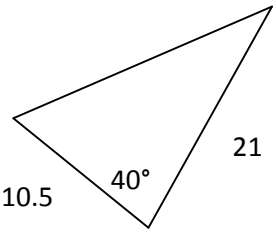




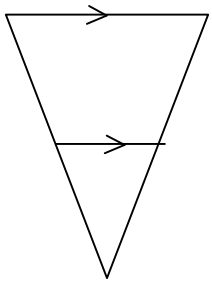
4) Of the following three triangles, which are similar and why?







5) Prove that the two triangles in the diagram are similar



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6) Prove that if two angles of a triangle are equal then the sides opposite those angles are equal

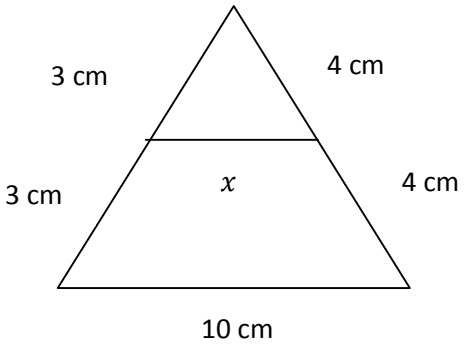
7) A tower casts a shadow of 40 metres, whilst a 4 metre pole nearby casts a shadow of 32 metres. How tall is the tower?

8) A pole casts a 4 metre shadow, whilst a man standing near the pole casts a shadow of 0.5 metres. If the man is 2 metres tall, how tall is the pole?

9) A ladder of length 1.2 metres reaches 4 metres up a wall when placed on a safe angle on the ground. How long should a ladder be if it needs to reach 10 metres up the wall, and be placed on the same safe angle?

10) A man stands 2.5 metres away from a camera lens, and the film is 1.25 centimetres from the lens (the film is behind the lens). If the man is 2 metres tall how tall is his image on the film?

11) What is the value of x in the following diagram?



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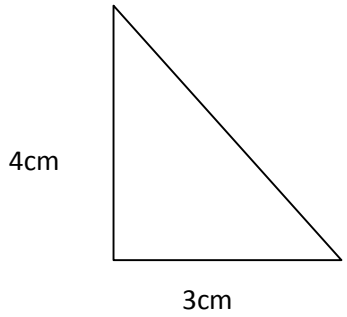
Exercise 4

Right Angled Triangles

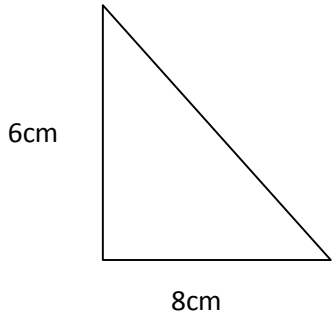
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1) Calculate the length of the hypotenuse in the following triangles

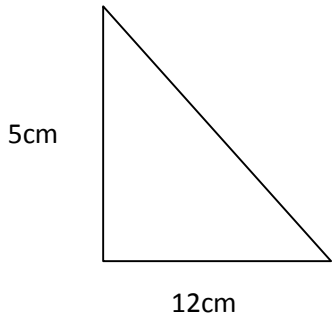
a)



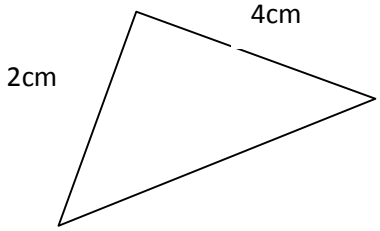
b)



c)

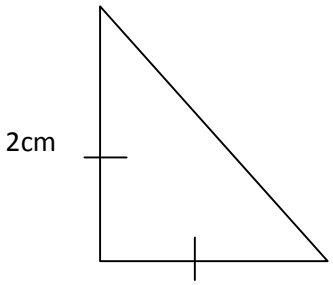


d)



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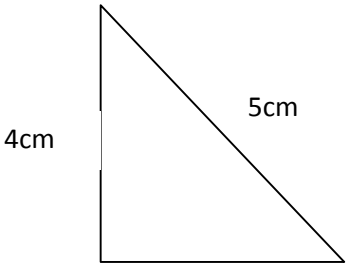
e)



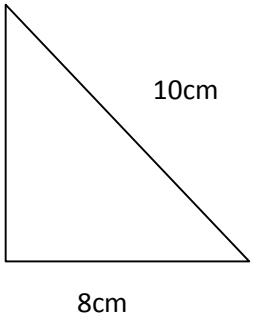
2) Explain why an equilateral triangle cannot be right-angled

3) Calculate the missing side length in the following triangles

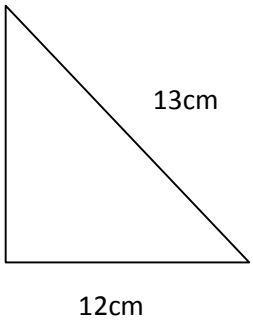
a)



b)

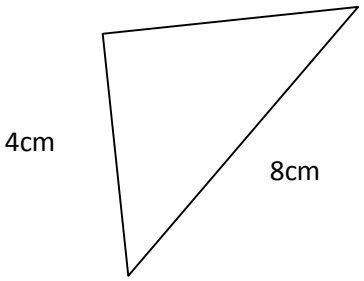


c)

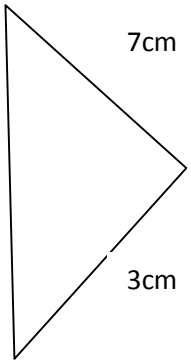


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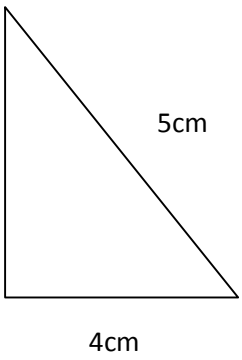
d)



e)



4) What is the area of the following triangle? (Use Pythagoras' to find required length)



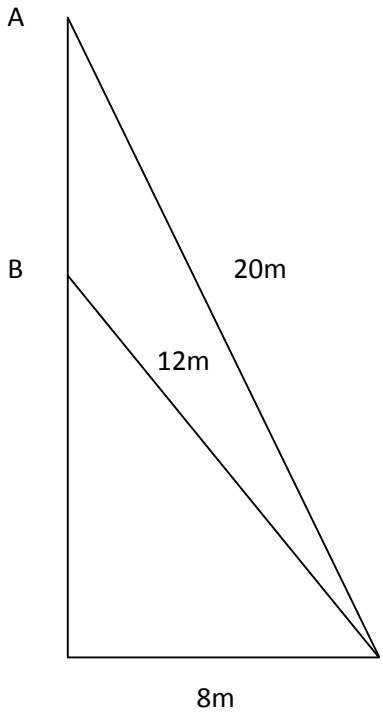
5) The equal sides of an isosceles right-angled triangle measure 8cm. What is the length of the third side?

6) A man stands at the base of a cliff which is 120 metres high. He sees a friend 100 metres away along the beach. What is the shortest distance from his friend to the top of the cliff?

7) A steel cable runs from the top of a building to a point on the street below which is 80 metres away from the bottom of the building. If the building is 40 metres high, how long is the steel cable?

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8) What is the distance from point A to point B?



9) A right angled triangle has an area of 20 cm^2. If its height is 4cm, what is the length of its hypotenuse?

10) What is the length of a diagonal of a square of side length 5cm?

11) A man is laying a slab for a shed. The shed is to be 6m wide and 8m long. To check if he has the corners as exactly right angles, what should the slab measure from corner to corner?

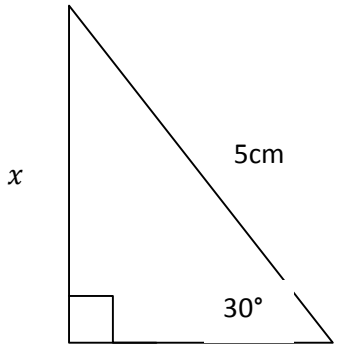
12) A box is in the shape of a cube. If the length of each side is 4cm, what is the length of a line drawn from the top left to the bottom right of the box?

13) The path around the outside of a rectangular park is 60m long and 40m wide. How much less will the walk from one corner of the park to another be if a path is built directly across the park from corner to corner?

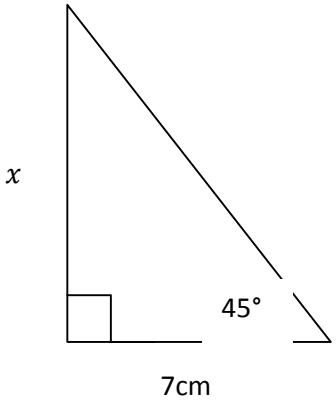
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14) Calculate the length of x in each of the diagrams below

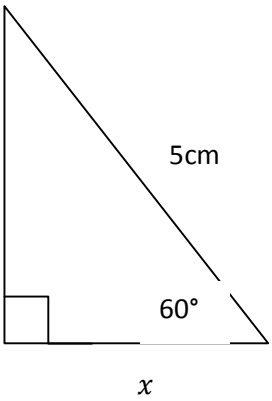
a)



b)

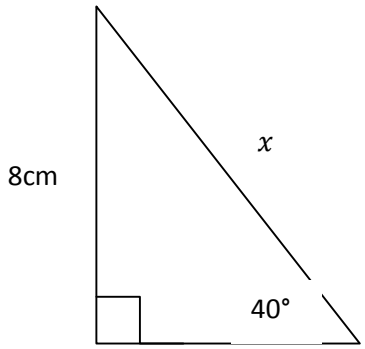


c)



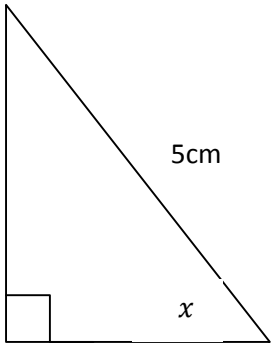
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d)



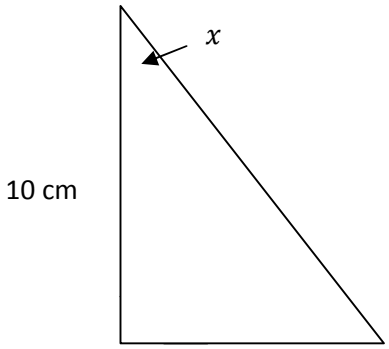
15) Calculate the size of angle x in the diagrams below, correct to the nearest degree.

a)



3 cm

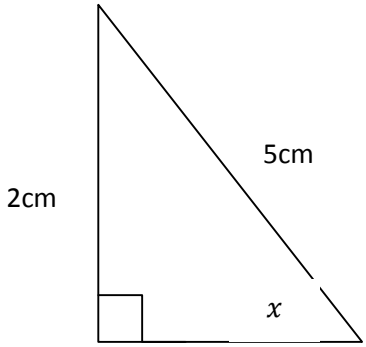
b)



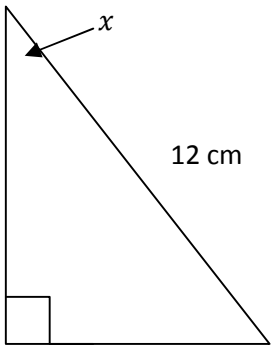
6cm

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c)

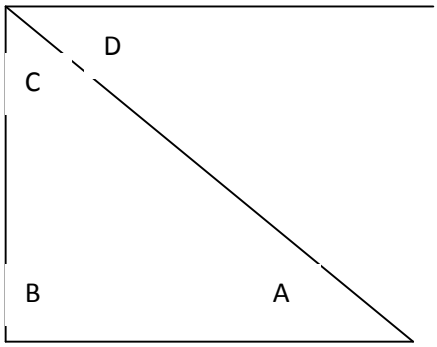


d)



6 cm

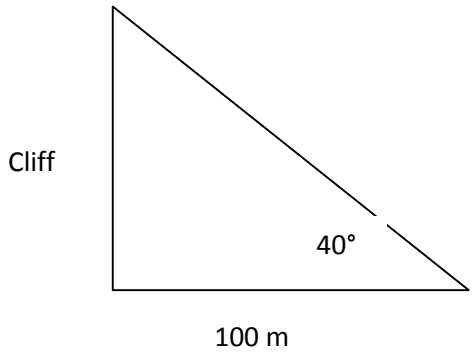
16) Identify the angles of elevation and depression in the diagram below



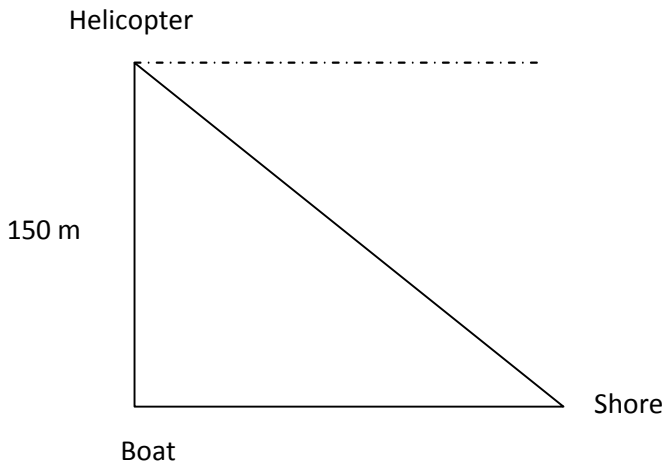
Complete the statement: The angle of elevation is ..... the angle of depression

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17) A man standing 100 metres away from the base of a cliff measures the angle of elevation to the top of the cliff to be 40 degrees. How high is the cliff?



18) A helicopter is hovering 150 metres above a boat in the ocean. From the helicopter, the angle of depression to the shore is measured to be 25 degrees. How far out to sea is the boat? (You need to fill in angle of depression on diagram)



19) A ramp is built to allow wheelchair access to a lift. If the angle of elevation to the lift is 2 degrees, and the bottom of the lift is 50 cm above the ground how long is the ramp?

20) The angle of elevation to the top of a tree is 15 degrees. If the tree is 10 metres tall how far away from the base of the tree is the observer?

21) From the top of a tower a man sees his friend on the ground at an angle of depression of 30 degrees. If his friend is 80 metres from the base of the tower how tall is the tower?

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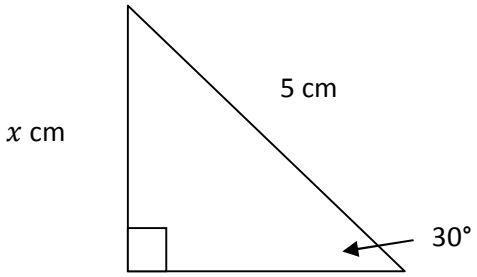
Exercise 5

Further Applications of Trigonometry

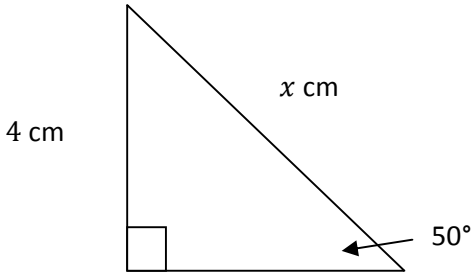
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1) Calculate the value of x in the following diagrams

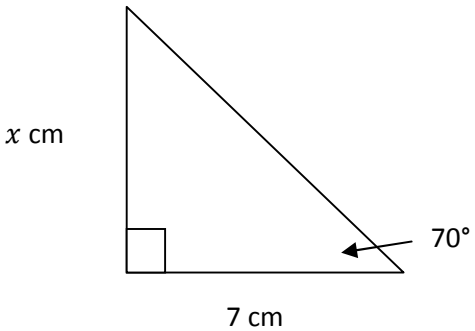
a)



b)

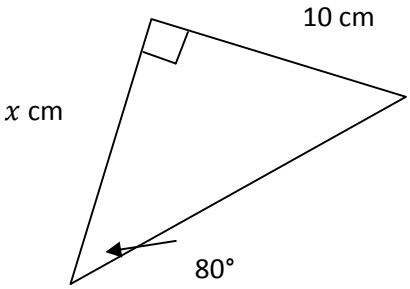


c)

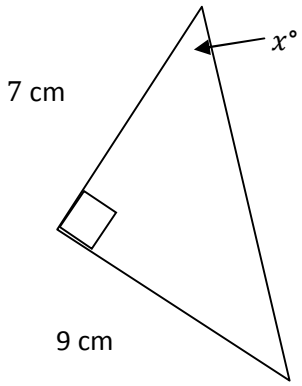


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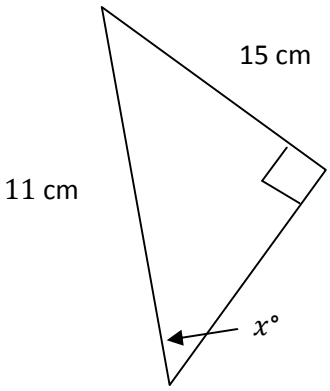
d)



e)



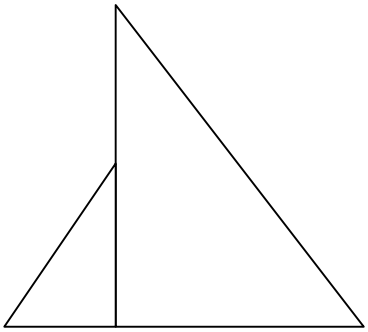
f)



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2) The foot of a ladder is 3 metres away from the base of a wall. If the ladder reaches 4.5 metres up the wall, what angle doe the foot of the ladder make with the ground?

3) Two sails sit back to back on a yacht. The first sail reaches half way up the second



The longest part of the second sail is 4 metres, and it makes an angle of 50 degrees to the deck. If the longest part of the first sail is 3 metres, what angle does it make with the deck?

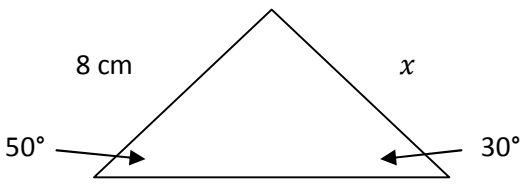
4) A piece of carpet is in the shape of a right angled triangle. The longest side is 80 cm, and it makes an angle of 65 degrees with the next side. What is the area of the piece of carpet?

5) Tom walks at an average speed of 4 km per hour in a north east direction. Ben walks at 5 km per hour, starting from the same point but in a south east direction. After 3 hours what is the shortest distance between them, and what is the angle from Tom to Ben?

6) A ship is on a bearing of 040 from a lighthouse, and a marker buoy is on a bearing of 310 from the same lighthouse. If the ship and the buoy are 100 km apart and the ship is 70 km from the lighthouse, what is the bearing of the buoy from the ship?

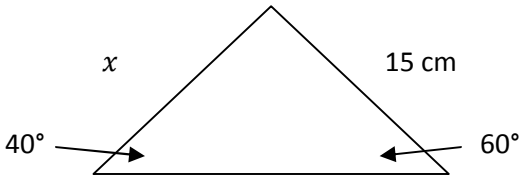
7) Calculate the value of x in the following diagrams

a)

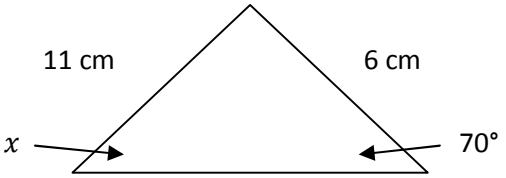


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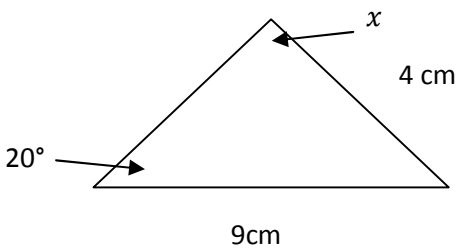
b)



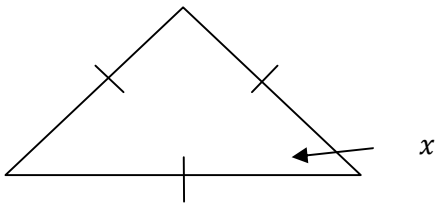
c)



d)



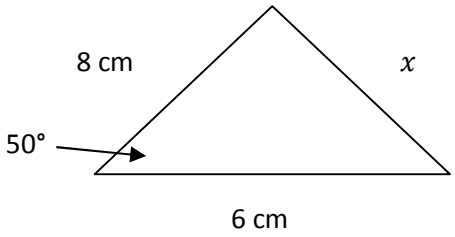
e)



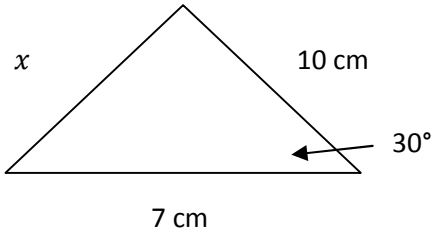
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8) Calculate the value of x in the following diagrams

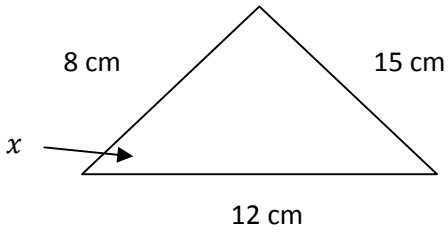
a)



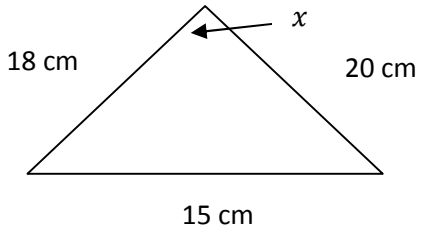
b)



c)



d)

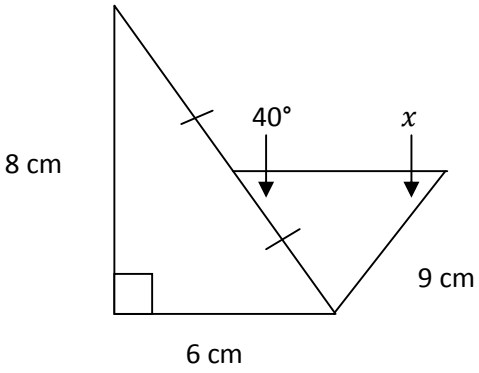


9) Calculate the area of each of the triangles in question 8

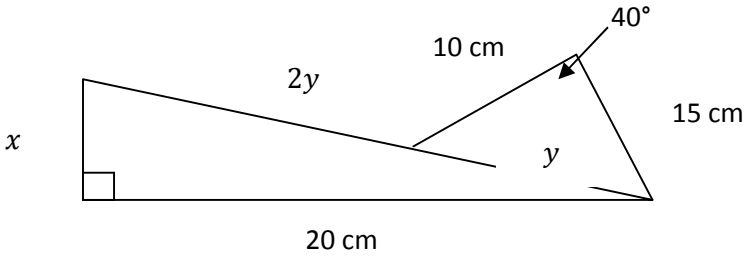
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10) Calculate the value of x in the following

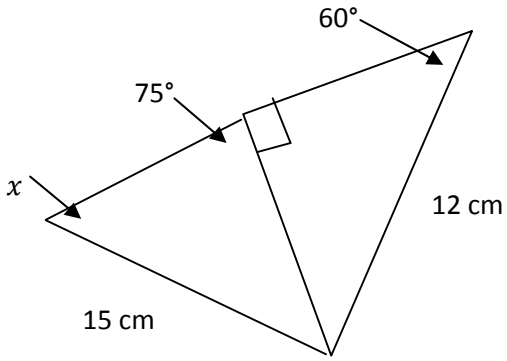
a)



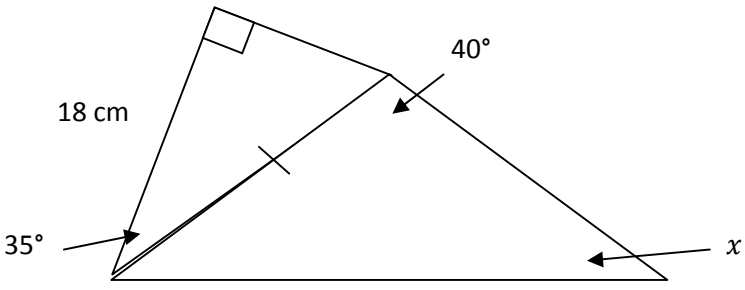
b)



c)



d)



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11) Thomas walks on a bearing of 15 degrees for 12 km, and Karl walks on a bearing of 125 degrees for 8 km. What is the shortest distance them after their walks?

12) Two wire ropes are attached to a tower; one on each side. The first rope makes an angle of 70° with the ground and is 15 metres long. If the second rope is 10 metres long, what angle does it make with the ground?

13) Three legs of a yacht race form a triangular course. The first leg is 10 km, and sails at some angle to the east of north the second is 8 km, and the third leg is 15 km. The start and finish points are the same. What angle is the first marker from the start point?

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Exercise 6

Spherical Geometry

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1) Complete the following table

Angle subtended by arc	Radius of circle	Arc length
90°	10 cm	
40°	25 cm	
70°		80 cm
125°		15 cm
	30 cm	90 cm
	90 cm	45 cm

2) State whether the following are true or false

a) All lines of latitude form great circles

b) Any two points on the same longitude form part of a great circle

c) Any two points on a sphere are parts of a circle

d) There is only one circle that can pass through 3 points on a sphere

e) The equator is a great circle

3) Find the latitude and longitude of the following cities to the nearest degree

a) Adelaide

b) Barcelona

c) Cairo

d) Jakarta

e) Lima

f) Mexico City

g) Osaka

h) Rome

i) Warsaw

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4) Convert the following to nautical miles

a) 1.852 km

b) 18.52 km

c) 312 km

d) 74.3 km

e) 1000 km

5) Convert the following to km

a) 1 nautical mile

b) 5 nautical miles

c) 0.1 nautical miles

d) 6.6 nautical miles

7) Calculate the shortest distances (in nautical miles and kilometres) between the following pairs of points (Assume Earth is a perfect sphere with a radius of 6400 km)

a) 26°N 40°W and 50°N 40°W

b) 10°N 30°E and 40°N 30°E

c) 45°N 25°W and 32°S 25°W

d) 9°N 75°W and 43°S 75°W

8) Calculate the time differences between the following cities using their longitudes (ignore daylight saving)

a) Athens and Adelaide

b) London and New York

c) Moscow and Anchorage,

d) Sydney and Nairobi

e) Bogota and Cairo

f) Tehran and Beijing

9) How much time would one gain or lose by flying between the following pairs of cities, given the flight time?

a) Cairo to Moscow takes 3 and a quarter hours

b) London to New York takes 6 and a quarter hours

c) Melbourne to Perth takes 3 hours

d) Paris to Tokyo takes 11 hours

e) Istanbul to New Delhi takes 5 hours

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Year 7 Mathematics

Probability

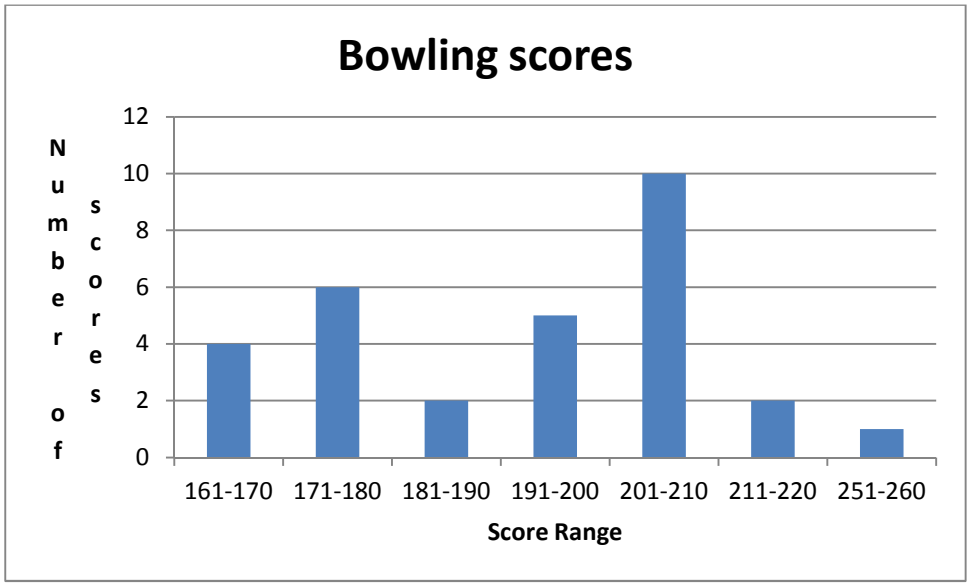
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Exercise 1

Simple Probability

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1) Peter plays ten pin bowling; his last 30 scores have been graphed in a frequency chart, shown here



Basing you answers on the chart data

a) Is Peter more likely to score 205 or 185 when he next bowls?

b) Is he more or less likely to score over 200 when he next bowls?

c) What would be his probability of scoring over 250 when next he bowls?

d) What would be his probability of scoring between 201 and 210 when next he bowls?

e) Discuss a major drawback with using this chart to predict the probabilities of future scores

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2) Craig rolled a pair of dice 360 times and recorded the sum of the two each time. He summarized his results in the table below

SUM of TWO DICE	Frequency
2	8
3	21
4	30
5	42
6	49
7	62
8	51
9	41
10	28
11	21
12	7

Based on his table:

a) What total is most likely to be rolled by two dice?

b) What is the most likely double?

c) What total is least likely to be rolled by two dice

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d) Is he more likely to roll a sum of 10 or a sum of 6 with two dice?

e) Is this data more reliable than that of Q1? Give two reasons to support your answer

3) What is the theoretical probability of each of the following?

a) A head being thrown when a coin is tossed

b) A blue sock being taken from a draw containing 3 blue and 5 red socks

c) The number 2 being rolled on a dice

d) An even number being rolled on a dice

4) A card is drawn from a standard pack of 52 cards. What is the probability of the card being:

a) A black card

b) A club

c) An ace

d) A black 2

e) A picture card

f) The 2 of diamonds

5) A man throws two coins into the air

a) List the possible combinations, and from this table:

b) What is the probability of throwing two heads?

c) What is the probability of throwing a head and a tail?

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d) If the first coin lands on a head, is the second coin more likely or less likely to be a head?

6) A coin is tossed and a dice is rolled

a) List the possible combinations of the coin and dice, and from this table:

b) What is the probability of throwing a six and a head?

c) What is the probability of throwing an odd number and a tail?

d) What is the probability of throwing a number higher than 4 and a head?

e) What is the probability of throwing a head and a 2 or a head and a 4?

7) A card is drawn from a normal pack. It is not replaced and a second card is drawn.

a) If the first card is red, what is the probability that the second card is also red?

b) If the first card is red, what is the probability that the second card is black?

c) If the first card is an ace, what is the probability that the second card is also an ace?

d) If the first card is the jack of clubs, what is the probability that the second card is the jack of clubs?

8) A set of cards consists of 10 red cards, numbered 1 to 10 and 10 black cards numbered 1 to 10

a) What is the probability of pulling a 10 at random?

b) What is the probability of pulling a black card at random?

c) What is the probability of pulling a red 2 at random?

d) What is the probability of pulling a red 2 on the second draw if the first card is a black 2, and it is not replaced?

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e) What is the probability of pulling an 8 on the second draw if the first card is an 8, and it is not replaced?

9) Consider the word ANATOMICALLY

a) What is the probability that a randomly chosen letter from this word will be an L?

b) What is the probability that a randomly chosen letter from this word will be an A?

c) What is the probability that a randomly chosen letter from this word will not be a vowel

d) What is the probability that a randomly chosen letter from this word will be a Z?

10) What is the probability that a digit chosen randomly from all digits (0- 9) is:

a) A prime number?

b) An even number?

c) Not 7?

d) Greater than 4?

e) Less than 10?

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Exercise 2

Multi-stage Events

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1) Construct a tree diagram that shows the possible outcomes of tossing a coin 3 times. List the sample space

2) Construct a tree diagram that shows the possible outcomes of rolling a four sided dice (numbered 1 to 4) twice. List the sample space

3) Peter has 3 green, 2 white and 4 black shirts in a draw. If he takes 3 out without replacing them construct a tree diagram that shows all possible outcomes and list the sample space

4) Repeat question 3, but assume Peter replace the shirt each time he pulls one out

5) For each of the above questions, relate the number of choices available for EACH event to the number of outcomes in the sample space

6) A man wants to visit three different towns; Alpha, Beta, and Gamma. If he can visit them in any order, but can only visit each town once per trip, how many different trips are possible? (List the possible trips)

7) In how many different ways can four separate coloured cards be arranged on a table?

8) From your answers to questions 6 and 7, establish a rule for determining the number of arrangements of any number of different objects. Use your rule to calculate the number of ways a man could read 5 books given that they can be read once only, but in any order

9) From a group of 4 people one is to wear a blue badge, and another a red badge. How many different combinations of people could wear the badges? (List the possibilities)

10) From a list of 5 books, John and Alex can choose one each. How many different combinations of books can they choose? (Note they cannot choose the same book as the other)

11) From your answers to questions 9 and 10, determine a rule for calculating how many different combinations of selections can be made from a list. Use your rule to determine how many groups of President, Secretary and Treasurer can be made from a committee of 5 people.

12) From a group of 4 people 2 are to be selected. How many different combinations are there?

13) From a group of 5 pizza toppings, a customer can choose two. How

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many different pizzas can be made?

14) From your answers to questions 12 and 13, determine a rule for selecting a certain quantity from a group. How does this differ from your answer to question 11? Use your result to calculate the number of groups of 3 people that can be chosen from a larger group of 8

15) Use tree diagrams to calculate the following probabilities

a) A coin is tossed three times and lands on heads each time

b) A four sided dice is rolled three times and the numbers 2, 4, and 1 are rolled (in any order)

c) A four sided dice is rolled and the numbers 2, 4, and 1 are rolled, in that order

16) A man pulls 3 shirts from a draw that initially contains 3 green, 2 white and 4 black shirts. If he does not replace the shirts what is the probability of drawing

a) One of each colour?

b) 2 black and 2 green?

c) All white?

d) All black?

17) Repeat question 16, but assume that the man replaces the shirts each time he pulls one out

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Exercise 3

Applications of Probability

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1) Tom tosses a coin four times in a row

a) How many different outcomes are there in each toss?

b) What is the probability of throwing a tail on any toss?

c) How many different outcomes are there for the four tosses?

d) Of these outcomes, how many times should Tom expect to throw four tails in a row?

e) Raise the probability of throwing a tail on one toss to the power of the number of tosses.

f) What do you notice about your answers to parts d and e?

g) Tom actually took 25 trials to throw four tails in a row. Does this mean the calculations are wrong? Explain your answer

2)

a) What is the probability of drawing a diamond from a standard pack of 52 cards?

b) What is the theoretical probability of drawing 5 diamonds in a row? (Assume the card is replaced each time)

c) Do you expect that an experiment would produce the exact result calculated in part b? Explain

3) Tim buys a ticket in a raffle which has three prizes. First receives $300, second gets $150 and third prize is $50. If there are 1000 tickets at $1 each, what is the financial expectation of Tim's ticket?

4) Glen always bets $5 on red at the roulette table. If the ball lands on red, Glen gets $10 back. If the ball lands on black, Glen loses his $5. If there are equal quantities of red and black numbers. What is Glen's financial expectation?

5) In reality there are also two green numbers on the wheel (0 and 00). If the ball lands on either of these, Glen (and every other player) loses. What is the new financial expectation?

6) Colin plays a game where there is a 30% chance of winning $4, a 20% chance of winning $10 and a 50% chance of losing $10. Each game costs 50 cents to play. What is his financial expectation?

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7) A group of 1000 people were asked whether smoking should be banned in restaurants, totally, only allowed in designated areas, or allowed anywhere in the restaurant. The results of the survey are shown in the following table

	Smokers	**Non-smokers**	**Total**
Banned	25	600	625
Special areas	75	100	175
Allowed	150	50	200
	250	750	1000

a) What is the probability that a person chosen at random wants smoking banned?

b) What is the probability that a smoker wants smoking banned?

c) What is the probability that a person who wants smoking to be allowed in special areas is a non-smoker?

d) What is the probability that a person who wants smoking banned is a non-smoker?

e) The surveyors claimed that the survey proves the majority of the population wants smoking banned in restaurants. How would you respond to this claim?

8) One thousand people take a lie detector test. Of 800 people known to be telling the truth, the lie detector indicates that 23 are lying. Of 200 people known to be lying, the lie detector indicates that 156 are lying. Present this information in a two-way table

9) A proposed test for a medical condition was trialled on 1000 volunteers, some who had the condition and some who did not. The trial was taken to determine how accurate the test was. The results are summarized in the table

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	Accurate	Not accurate	Total
With condition	195	5	200
Without condition	730	70	800
Total	925	75	1000

a) Why were only 200 people with the condition included in the trial of 1000 people?

b) What was the overall correct diagnosis percentage?

c) What is the probability that a person with the condition is properly diagnosed?

d) What is the probability that a person who did not have the condition was incorrectly diagnosed (that is told they had the condition)?

e) What is the probability that a person who was diagnosed incorrectly did not have the condition?

f) Comment on the overall effectiveness of the test

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General Mathematics

Algebraic Modelling

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Exercise 1

Algebraic Skills & Techniques

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1) Substitute the value $x = 2$ into each of the following linear equations, and hence evaluate the equation

a) $x + 2$

b) $x + 3$

c) $x - 5$

d) $2x + 1$

e) $3x - 4$

f) $6x + 5$

g) $2x - 10$

h) $3x - 6$

i) $\frac{1}{2}x + 8$

j) $\frac{1}{4}x - 4$

k) $\frac{7}{2}x - 7$

l) $0x + 23456$

2) Substitute the value $x = 3$ into each of the following quadratic equations, and hence evaluate the equation

a) x^2

b) $x^2 + 3$

c) $x^2 - 2$

d) $2x^2 + 2$

e) $3x^2 - 1$

f) $5x^2 + 2$

g) $2x^2 - 20$

h) $4x^2 - 30$

i) $\frac{1}{2}x^2 + 1$

j) $\frac{1}{3}x^2 - 3$

k) $\frac{2}{3}x^2 + 5$

l) $0x^2 + 21232$

3) Substitute the value $x = 2$ into each of the following cubic equations, and hence evaluate the equation

a) x^3

b) $x^3 + 1$

c) $x^3 - 2$

d) $2x^3$

e) $3x^3 - 24$

f) $\frac{1}{2}x^3 - 6$

g) $0x^3 + 123432$

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4) Simplify the following expressions

a) $4x - 2x$

b) $3x + x$

c) $3x + y - x$

d) $4x + 2y - x + y$

e) $3x - 2y + 3y - 4x$

f) $2x^2 - 3x + x^2 - 2x$

g) $x - x^2 + y - x^2 - y$

h) $2x^2 - y + y^2 - y - 2x^2$

i) $-y + 3x - 2x^2 - 3x + y$

j) $(2x + y) - (y - 2x)$

k) $(y - 3x) + (3x^2 - y)$

5) Multiply the following, expressing your answer in index form

a) $2x \times y$

b) $3x \times x$

c) $3x^2 \times x^2$

d) $y^2 \times x \times y$

e) $x \times y \times z \times y^2$

f) $a \times b^2 \times a^2 \times b$

6) Simplify the following

a) $\frac{3x^2}{x}$

b) $\frac{4xy}{2y}$

c) $\frac{8x^3y^2}{2xy}$

d) $\frac{15xy^3z}{5xyz}$

e) $\frac{x^4y}{x^2y^2}$

f) $\frac{x \times x^2 \times z^2 xz^2}{x^3 \times z^4}$

7) Make t the subject of the following equations

a) $y = 3t - 1$

b) $y = \frac{1}{2}t + 2$

c) $2y = 3t - 5$

d) $2y - 3 = 2t + 4$

e) $y = t^2$

f) $y = \frac{1}{2}t^2$

g) $y = x^2 - t^2$

h) $y = at^2 + c$

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8) Solve for y by substituting the value given into the equation

a) $y = 3x - 2$when $x = 3$

b) $y = 7x - 7$ when $x = 1$

c) $y = \frac{1}{2}x^2 - 4$ when $x = 4$

d) $y = \sqrt{x} - 3$ when $x = 16$

e) $y = 2x^2 + 3x - 1$ when $x = 3$

f) $y = 3(x^2 - 100)$ when $x = 20$

g) $y = \frac{100}{x} + x^2$ when $x = 15$

h) $x = 3y^2$ when $x = 27$

i) $x = \frac{1}{2}y^2 - 2$ when $x = 70$

j) $x = \frac{2y-4}{3}$ when $x = 6$

9) Express the following in scientific notation

a) 0.0356

b) 21223.19

c) 409.754

d) 0.00787

e) 19003

f) 32.856

g) 0.00342

h) 499.005

10) Use guess check and improve to calculate the value of x in the following

a) $3^x = 12$

b) $2^x = 14$

c) $2^{x-1} = 9$

d) $0.5^{2x} = 0.25$

e) $10^{4x} = 100$

f) $1^x = 700$

g) $34^x = 1$

11) A tree loses 20% of the leaves it has each day. After how many days will it have 10% of its original number of leaves left?

12) A balloon is blown up so its size increases by 25% each minute. It bursts after 8 minutes. How much bigger than its original size was it when it burst?

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13) The total resistance of two resistors placed in parallel in an electrical circuit is given by the formula

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

Where R is the total resistance in the circuit, and R_1 and R_2 are the values of the two resistors

If the value of R_1 is fixed at 10 ohms, draw up a table of values for R when R_2 is 5, 10, 15, ...50 ohms

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Exercise 2

Modelling Linear Relationships

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1) For each of the linear functions, draw a table of values for $-3 \leq x \leq 3$ and sketch the graph of the function from your table

a) $y = x$

b) $y = 2x + 2$

c) $y = 3x - 4$

d) $y = \frac{1}{2}x + 2$

e) $y = -x - 1$

f) $y = -2x + 5$

g) $y = -\frac{1}{2}x + 4$

2) From your answers to question 1, what is the relationship between the value of the constant in a linear equation, and the graph of the equation?

3) From your answers to question 1, what is the effect of changing the sign of the coefficient of x?

4) Choose two pairs of graphs from question 1 and determine their point(s) of intersection

5) The instructions for cooking a roast state that it should be cooked for thirty minutes plus 40 minutes for every kg the meat weighs

a) For how long should a roast that weighs 1.5 kg be cooked for?

b) Construct a table of values that relate the weight of the meat to its cooking time

c) Graph the values

d) Determine the gradient of the line produced. How does this value relate to the quantities in the problem?

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e) Relate the y intercept to the quantities in the problem

f) Is the graph valid for all weights; that is can the graph be extended indefinitely? Explain your answer

6) A plumber charges a call out fee of $25 plus $20 per hour for his work. If he works for part of the hour he only charges for that part. For example, for 15 minutes work he will charge $5 (plus his call out fee)

a) How much will he charge for 2 hours work?

b) How much will he charge for 3.5 hours work

c) Construct a table of values that relate the time taken for a job to the total charge

d) Graph the values

e) Determine the gradient of the line produced. How does this value relate to the quantities in the problem

f) Relate the y intercept to the quantities in the problem

g) Is the graph valid for all weights; that is can the graph be extended indefinitely? Explain your answer

7) Another plumber charges a $25 call out fee and $20 per hour for his work. Differently to the previous plumber he charges $20 even if he only works for part of an hour. For example, for 15 minutes work he will charge $20 (plus his call out fee)

a) How much will he charge for 2 hours work?

b) How much will he charge for 3.5 hours work

c) Construct a table of values that relate the time taken for a job to the total charge

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d) Graph the values

e) How does the graph differ from that in question 6?

8) To convert from Celsius to Fahrenheit temperature the following formula is used

$$F = \frac{9}{5}C + 32$$

a) Construct a table of values for $C = 0\ to\ 40$ in steps of 5 degrees

b) Graph the relationship

c) Determine the gradient of the line produced. How does this value relate to the quantities in the equation?

d) Relate the y intercept to the quantities in the equation

e) Use the graph to extrapolate the value of 42 degrees Celsius in Fahrenheit

f) Use the graph to determine how many degrees Celsius equals 23 degrees Fahrenheit

g) Is the graph valid for all values of C? Explain

9) One Australian dollar currently buys 56.5 Indian rupees

a) Construct a table of values for 0 to 30 Australian dollars in steps of 5 dollars

b) Graph the relationship

c) Determine the gradient of the line produced. How does this value relate to the quantities in the equation?

d) Relate the y intercept to the quantities in the equation

e) How many rupees does 40 Australian dollars buy?

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f) How many Australian dollars does 1695 rupees buy?

10) A bath has 200 litres of water in it. The plug is pulled and water flows from it at the rate of 4 litres per second.

a) Construct a table of values that relate the volume of water in the bath to the time since the plug was pulled

b) Graph the relationship

c) From your graph how long until the bath is empty?

d) Determine the gradient of the line produced. How does this value relate to the quantities in the problem?

e) Relate the y intercept to the quantities in the problem

f) Is the graph valid for all values of t? Explain

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Exercise 3

Modelling Non-linear Relationships

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1) For each of the following equations, generate a table of values for $-3 \leq x \leq 3$ and sketch the graph of the function from your table

a) $y = x^2$

b) $y = x^2 - 2$

c) $y = x^2 - 1$

d) $y = 2x^2$

e) $y = -x^2 + 1$

f) $y = -2x^2 - 2$

g) $y = -\frac{1}{2}x^2 + 4$

2) From your answers to question 1, what is the effect of changing the sign and value of the coefficient of x in a quadratic equation?

3) From your answers to question 1, what is the relationship between the value of the constant in a quadratic equation and the graph of the equation?

4) Using your graphs, find the co-ordinates of the maximum or minimum values of each function in question 1

5) Make a table of values for each pair of equations

a) $(x-1)^2 + 3, \quad x^2 - 2x + 4$

b) $(x+2)^2 - 1, \quad x^2 + 4x + 3$

c) $(x-2)^2 + 2, \quad x^2 - 4x + 6$

d) $(x+1)^2 + 1, \quad x^2 + 2x + 2$

e) $(x-3)^2 - 6, \quad x^2 - 6x + 3$

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6) What do you notice about the table of values for each pair of equations in question 5, and hence their graphs?

7) What can you say about each pair of equations in question 5?

8) For each equation, generate a table of values and graph the equation, choosing an appropriate range

a) $y = x^3$

b) $y = 2x^3$

c) $y = -x^3$

d) $y = -3x^3$

9) For each equation, generate a table of values and graph the equation, choosing an appropriate range

a) $y = 2^x$

b) $y = \left(\frac{1}{2}\right)^x$

c) $y = \left(\frac{1}{4}\right)^x$

d) $y = 3^x$

10) How is the graph of the equations in question 13 different for $a > 1$ or $a < 1$

11) For each equation, generate a table of values and graph the equation, choosing an appropriate range

a) $y = \frac{1}{x}$

b) $y = \frac{2}{x}$

c) $y = \frac{4}{x}$

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d) $y = \frac{-1}{x}$

12) The distance an object falls due to gravity on Earth can be approximated by the equation $d = 5t^2$, where d is the distance in metres, and t is the number of seconds. Graph this equation, and use it or a table of values to determine

a) How far an object falls in 5 seconds

b) The time an object has been falling if it gas travelled 80 metres

13) On the moon gravity is weaker, so whilst the equation from question 16 still generally applies, the coefficient is different. After 2 seconds on the moon an object has fallen 3.2 metres.

a) Calculate the new coefficient and hence the equation describing the distance a body falls in t seconds on the moon

b) How far has a body on the moon fallen after 10 seconds?

c) A body falls 28.8 metres on the moon. How long has it been falling for?

14) An ant is removing small rocks from a pile. The number of rocks left in the pile can be approximated by the equation $N = \frac{N_0}{t+1}$ where N is the number of rocks remaining, t is the time in minutes, and N_0 is the number of rocks initially in the pile. After 3 minutes there were 25 rocks in the pile

a) How many rocks were in the pile initially?

b) How many rocks had the ant removed after 1 minute?

c) How many rocks will remain after 9 minutes?

d) Explain why this equation can only be considered as an approximation. (Hint look at large values of t)

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15) Water flows from a large hose at the rate of 16 litres per minute. At this rate it takes 22 hours to fill a small pond. If the flow rate reduces to 4 litres per minute, it takes 88 hours to fill the pond

a) Calculate the proportionality constant for this situation, and hence produce the equation relating the flow rate to the time taken to fill the pond

b) How many litres does the pond hold?

c) How long would it take to fill the pond if the flow rate was changed to 32 litres per minute?

d) If it took 11 hours to fill the pond, what was the flow rate?

16) John deposits $10000 into a bank account that pays $x\%$ interest compounded annually. He deposits no other funds, and after 3 years his balance is $12597.12.

a) Calculate the interest rate, and hence write the equation that relates John's balance after t years

b) What will John's balance be after 10 years?

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